Trace-Trace Crosstalk

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Introduction

This lab aims to demonstrate the effect of the return path of a signal on the switching noise measured on parallel traces. To accomplish this, the lab features a PCB with two sections, one with a common ground return plane and one with a toggleable separate return trace or common return trace. Each trace on this board is connected to an LED.

To demonstrate the effect of the return path on noise, the following objectives were employed.

NOTE: A Whadda development board was used, however, since it is identical to an Arduino UNO, it will be referred to as an Arduino.

1. Generate a fast-switching pulse signal on pins 8 to 13 of an Arduino.
2. Test different modes of pulsing to determine if features other than pulse speed affects noise.
    a. All pins switching simultaneously.
    b. Each pin switching one at a time.
    c. An increasing number of pins switching.
3. Read the voltage signature on the victim loop to observe the changes in voltage.

Plan of Record

What does it mean to work?

As the board has already been constructed, the POR will only involve the signature read on the victim line.

1. The Arduino (or Arduino equivalent) must generate a pulse signal with a rise time of under 1 us.
2. All LEDs must be on with no visible blinking.
3. There must be a noticeable change in the voltage signature of the victim loop.
4. Each mode of switching must generate a different voltage change.

Features of the Board:

1. 2-layer stack board, with only the bottom half of the board containing a ground plane.
2. Top half of the board identical signal traces with either a single common return trace (sharing a path with the victim loop) or a return trace separate from the victim loop.

Components:

NOTE: All components will listed are for the testing of the board and have nothing to do with the creation of the Lab 9 board.

1. Arduino
2. 10x Probes x2
3. Lab 9 Board

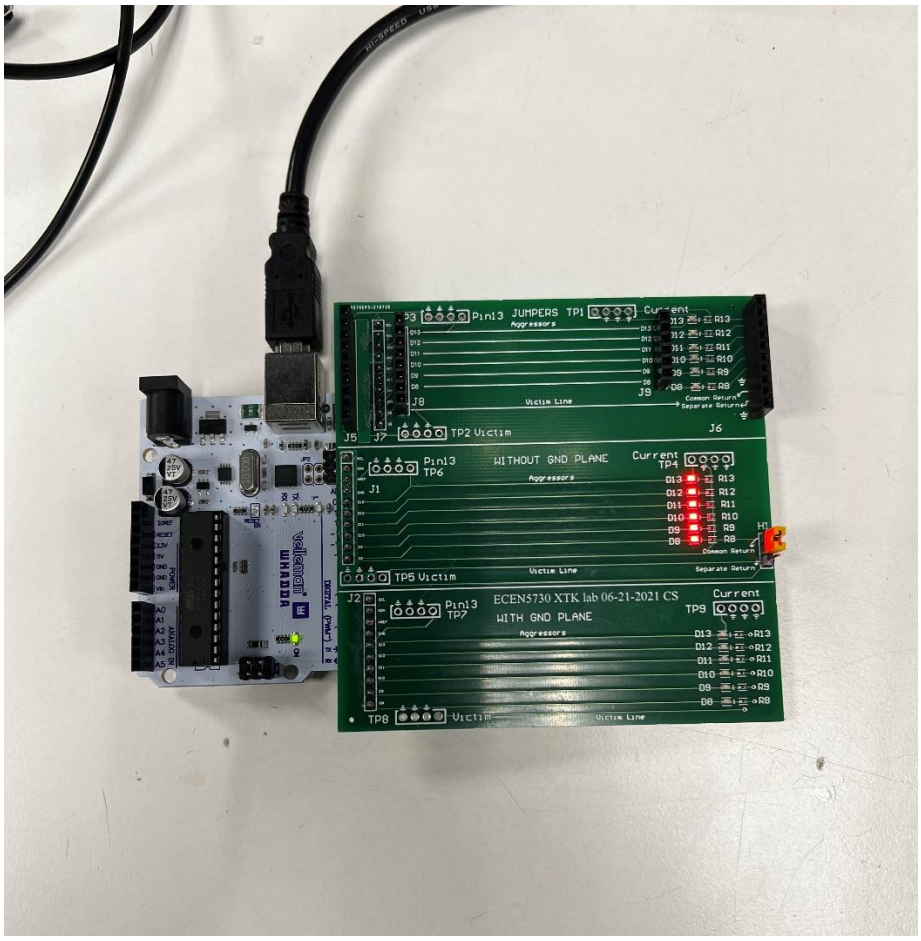


Fig 0: Lab 9 Board

The top third of the board is not relevant to the lab. Moving forward, the lab 9 board will only refer to the bottom and middle third of the board in figure 0.

Arduino Code For Simultaneously Switching Pulse Waves:

```
void setup() {
pinMode(13, OUTPUT);
pinMode(12, OUTPUT);
pinMode(11, OUTPUT);
pinMode(10, OUTPUT);
pinMode(9, OUTPUT);
pinMode(8, OUTPUT);
}
void loop() {
PORTB=B00111111;
PORTB=B00000000;
}

/*
  ID for pins 8 to 13:
  8: B00000001
  9: B00000010
  10: B00000100
  11: B00001000
  12: B00010000
  13: B00100000
 */
```

Fig 1: Code used for the pulsing signal generated by the Arduino

Figure 1 shows the main code used to generate the pulse signal. The PORTB command was used rather than a write command since we need the signal to switch simultaneously. The write commands are not able to generate a simultaneously switching signal, which would skew any conclusions made from the victim loop voltage signature.

To generate the modes of pulsing, the byte value of the pin we want to switch was activated (changed to 1) while the other values remain 0.

Characteristics of the Pulse Wave:

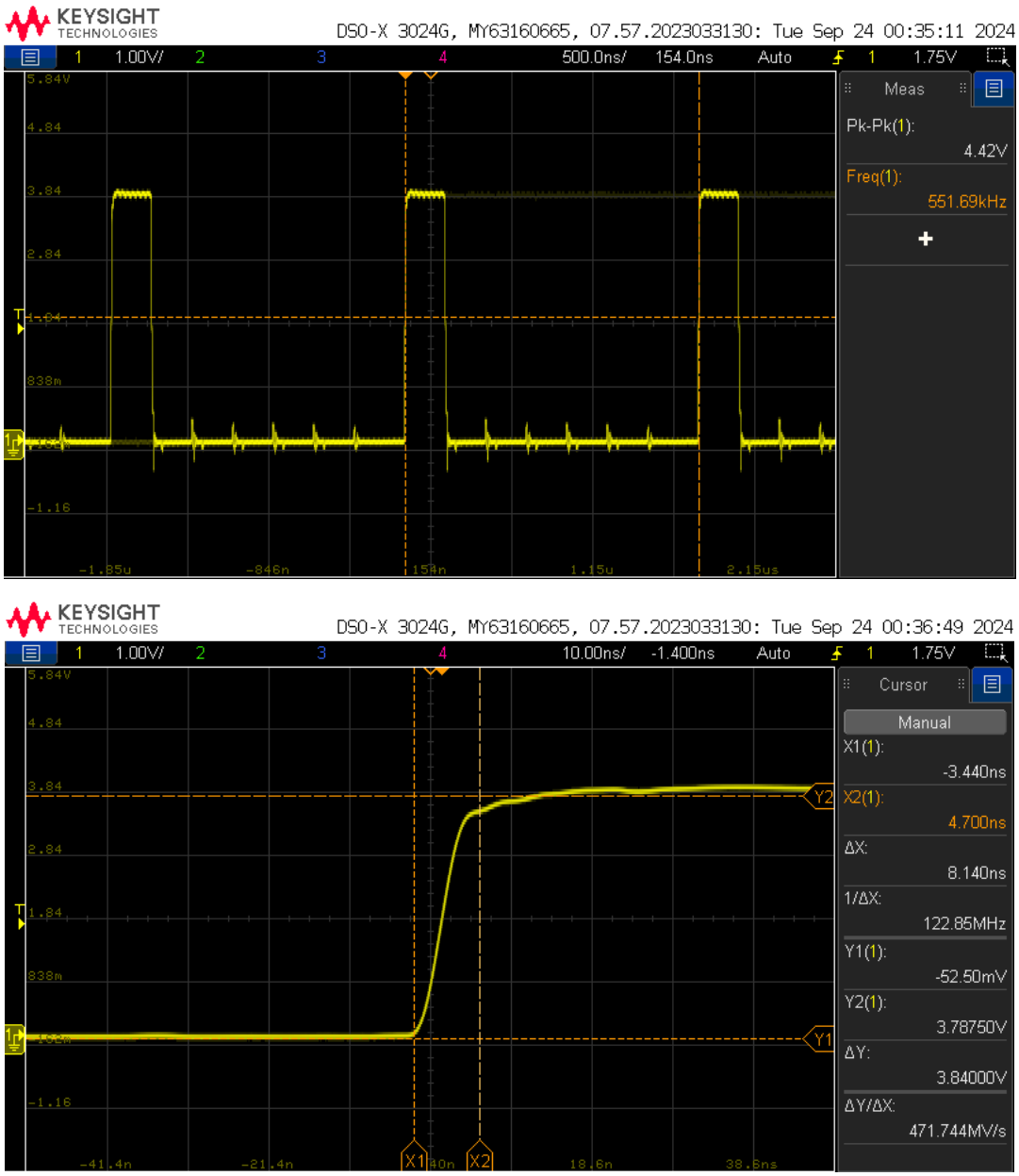


Fig 2: Overall waveform (top) and pulse risetime (bottom)

Figure 2 displays the waveform generated on the pins of the Arduino. Here, it is shown that the Arduino can generate a signal with a rise time of 8.7 ns with a peak-peak amplitude of 4.42v, which verifies that this signal will generate ample noise on the victim line. It should be noted that p-p amplitude is important as that will generate a measurable voltage drop on the victim loop. From lab 5, we learned that the equation for the induced inductance of a long trace is dependent on the voltage drop on the trace and the speed of the switching (i.e. change in current). A high p-p voltage implies a large change in current which allows for a calculation for the inductance of that trace.

It must also be noted that the scope is triggered on the pulse generated on pin 13. Due to this, pin 13 will be switching in every scope capture and will be influencing the noise on the victim line. This is a design flaw of the board as we cannot separate pin 13 from the rest of the switching signals.

Part 1: Common Return Plane

The first section of the board that will be analyzed is the bottom half of the board which features a common return plane for all signal traces.

All Pins Switching Simultaneously:

For this mode, all pins were allowed to switch simultaneously.

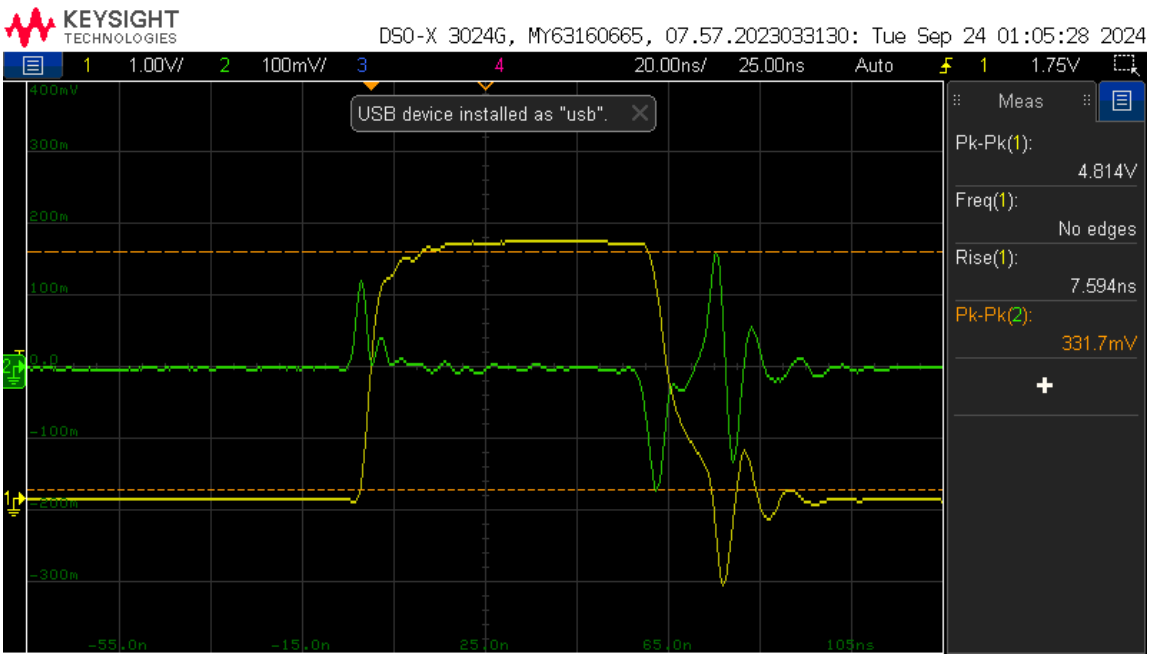


Fig 3: Switching noise (green) with all pins switching simultaneously

NOTE: The scales on the victim trace and pin trace are different, for the rest of this part, all figures will feature the victim loop at 100mV per division and the aggressor at 1V per division.

In figure 3, there is a significant fluctuation in voltage on the victim loop when the pin switches.

Pins Switching One at a Time:

NOTE: For the sake of easy comparison and space, the graphs have been shrunken down, full graphs have been included in the index.

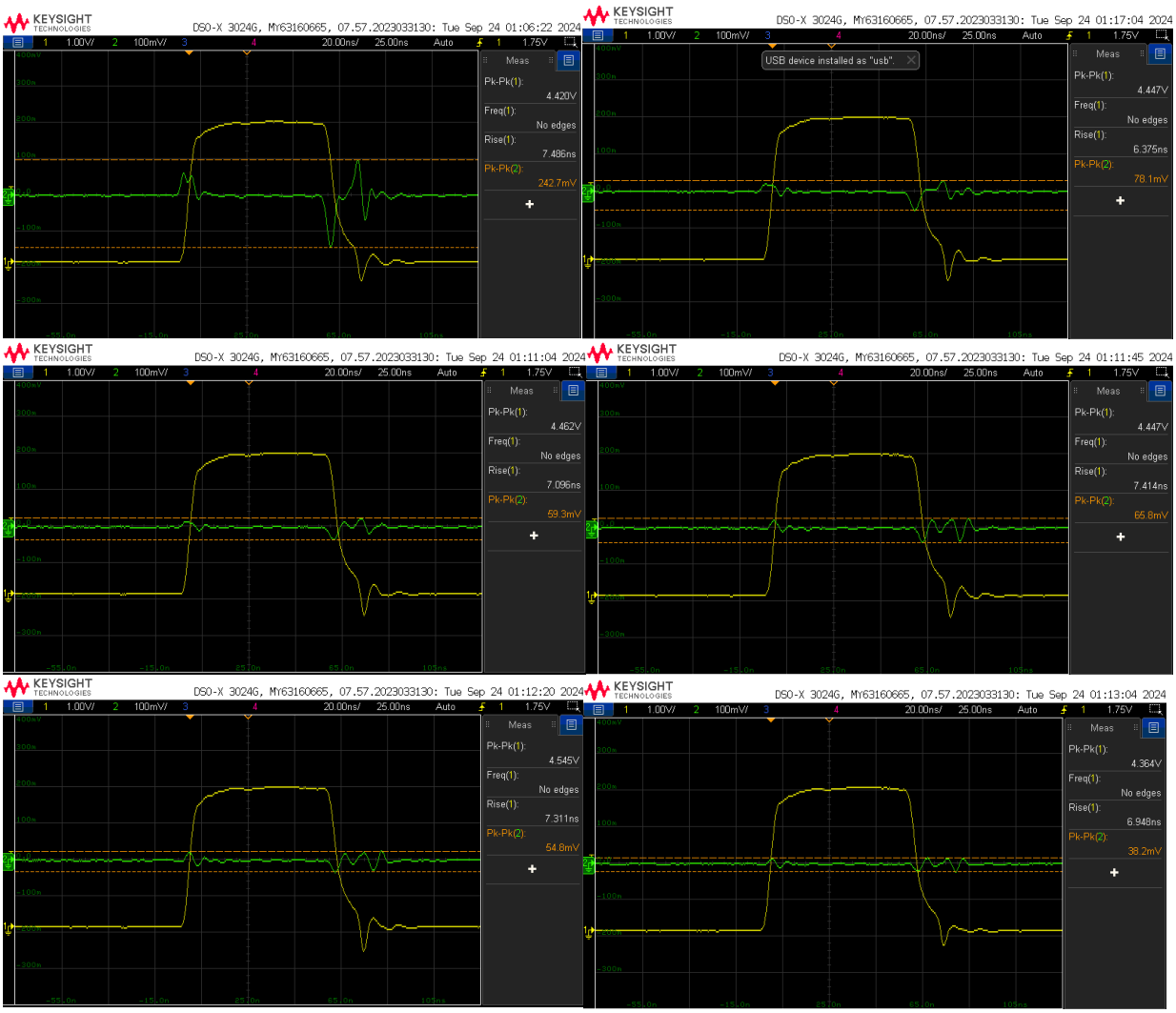


Fig 4: Switching noise when pin 8 (top left) 9 (top right) 10 (middle left) 11 (middle right) 12 (bottom left) and 12 (bottom right) switch

Increasing Number of Pins Switching:

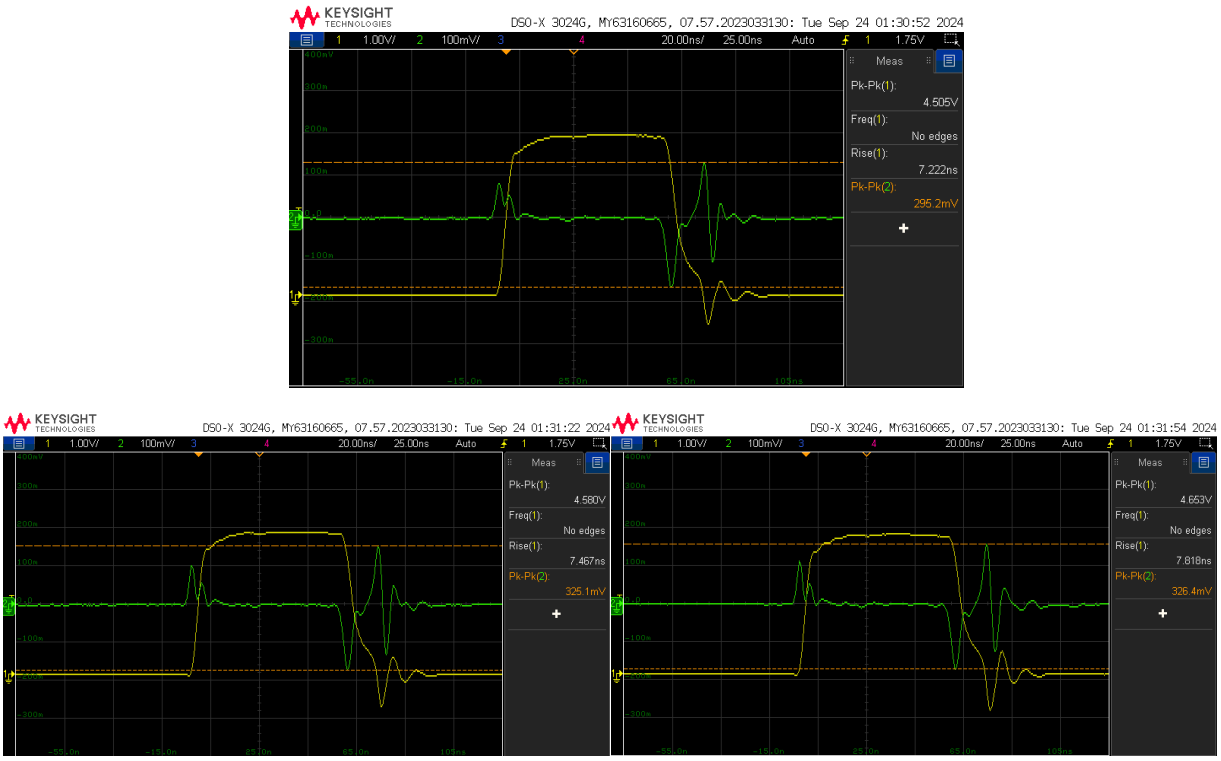


Fig 5: Switching noise from pins 8 and 9 (top) 8, 9 and 10 (bottom left) 8, 9, 10 and 11 switching simultaneously.

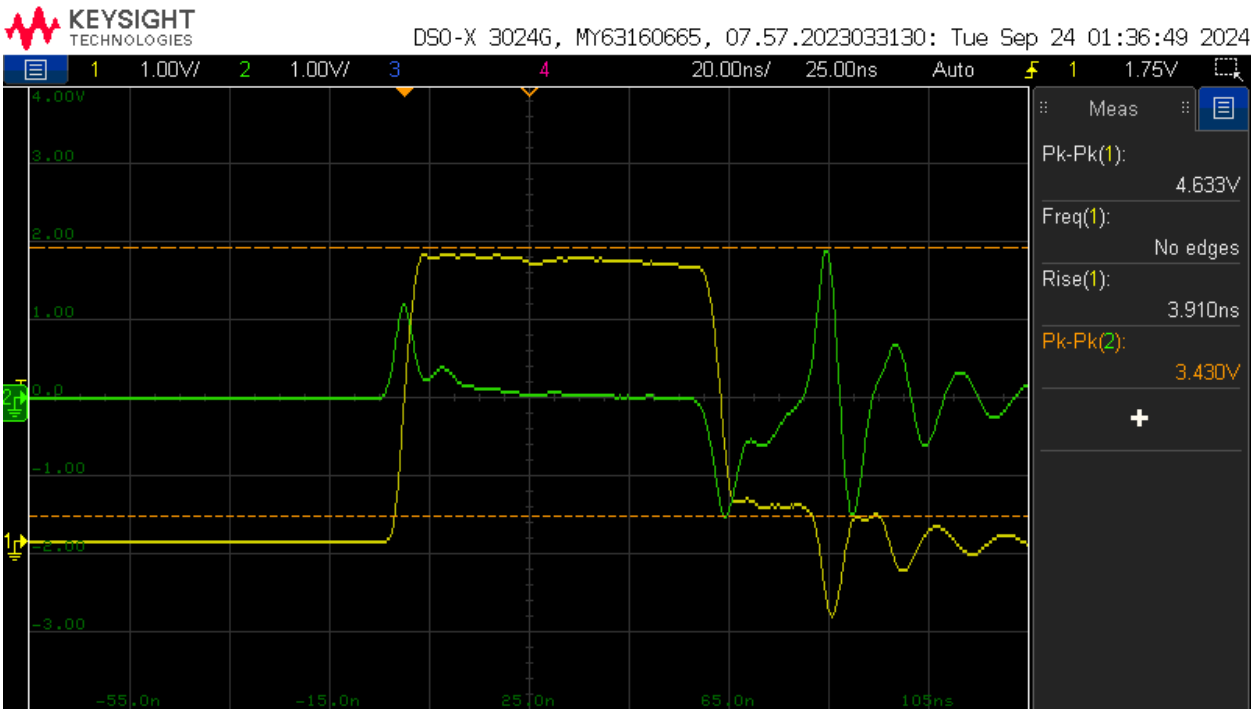
From figures 3, 4 and 5, we can deduce the behavior of the victim loop under different switching circumstances. In figure 4 we see that pin 8 dominates and causes the most noise on the victim loop than the other traces and in figure 5, adding pins 11 and 12 does not seem to cause a significant increase in noise.

From these findings, it is evident that with a ground plane, if traces are closer together there is a larger amount of cross-talk between them than if they were further apart. Long, parallel signal traces should be avoided for best design practices.

Part 2: Common/ Shared Return Path

To visualize the difference between using a common return path vs. a separate one, all pins were allowed to switch and the voltage signature on the victim loop was read in both situations.

All Pins Switching Simultaneously:



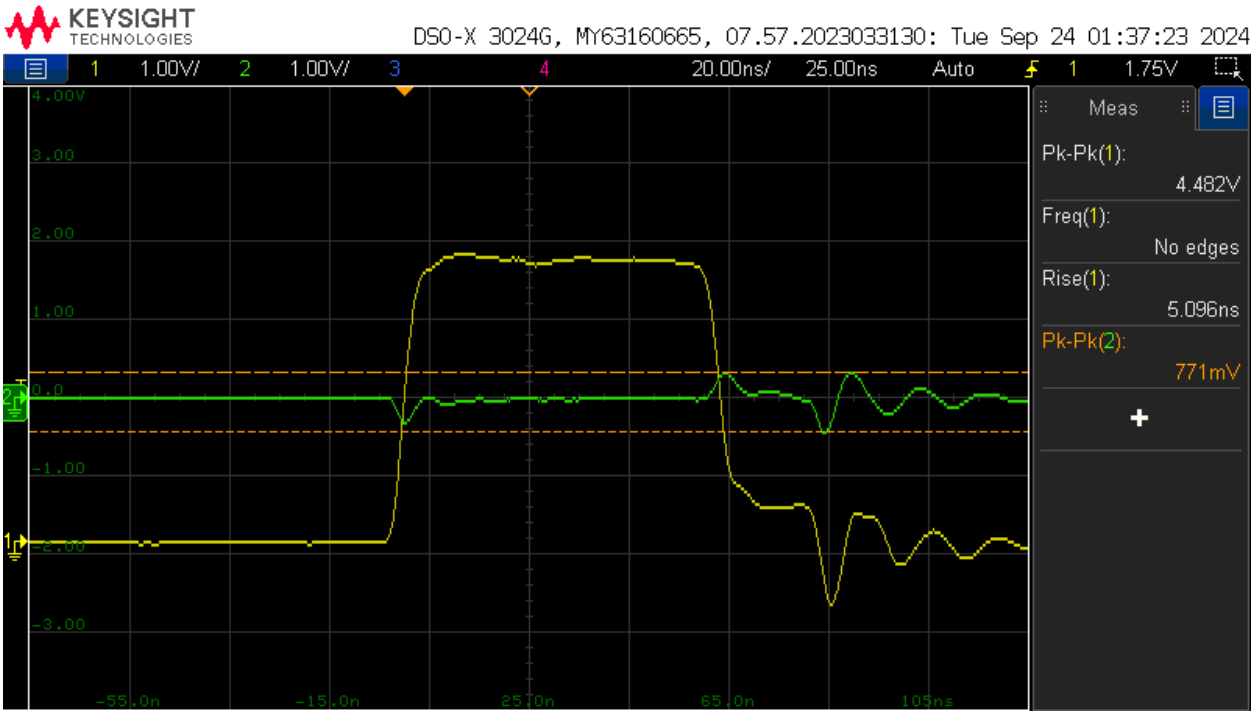


Fig 6: Switching noise with a common return path (top) and separate return (bottom)

It is important to immediately note that both graphs are displayed at the same scale, indicating a much larger degree of noise on the victim loop than with the ground plane. Additionally, the noise experienced with a common return path is much greater with a voltage change of 3.43V than with a separate path with a change of 771 mV. It is safe to assume that a common path will always cause more cross-talk between

traces, therefore, it is sufficient to show the effect of different pins switching with just the cross-talk from separate paths.

One Pin Switching at a Time (Separate Path):

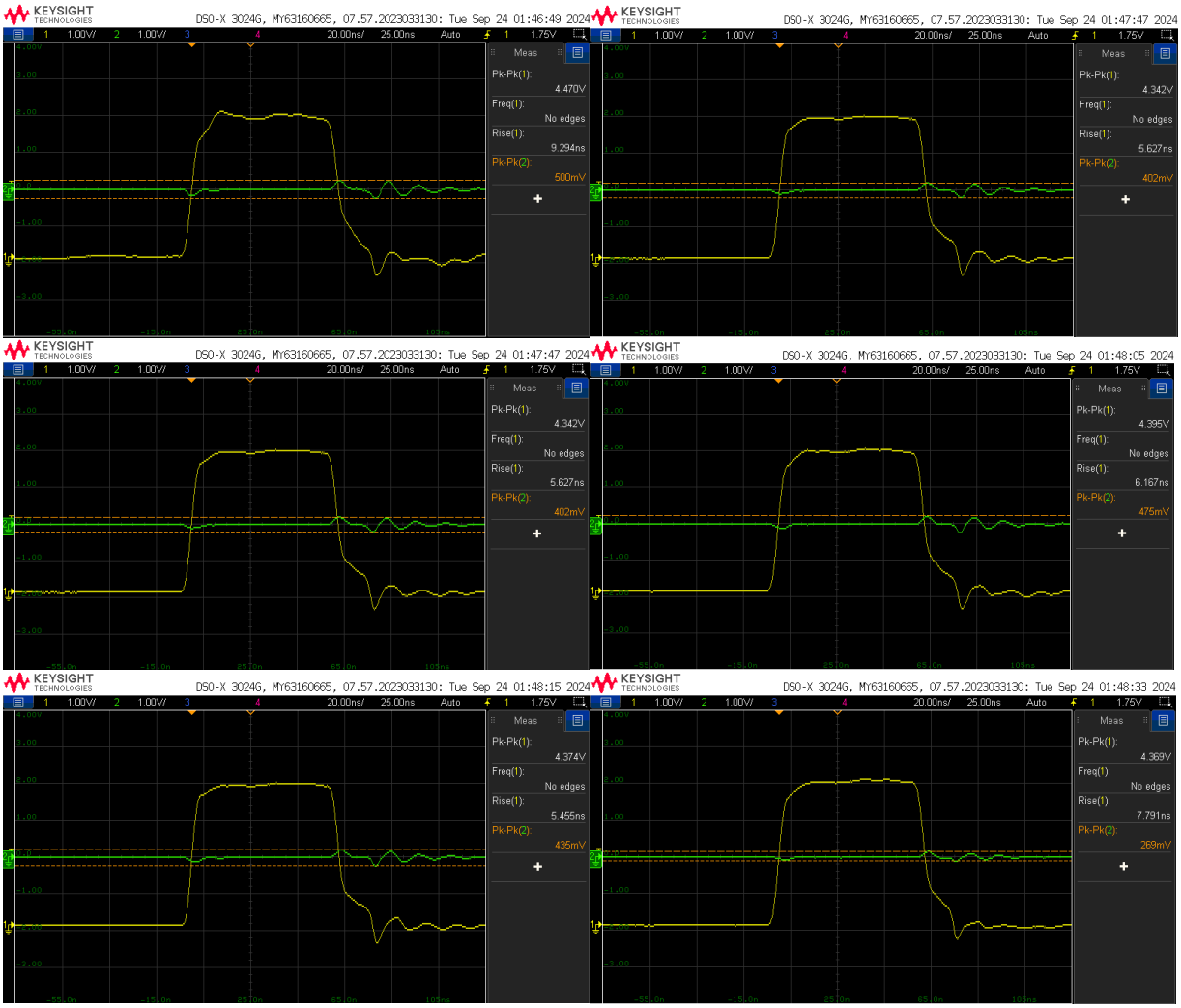


Fig 7: Switching noise on a separate return path when pin 8 (top left), 9 (top right), 10 (middle left), 11 (middle right), 12 (bottom left) and 13 (bottom right)

Increasing Number of Pins Switching (Separate Path):

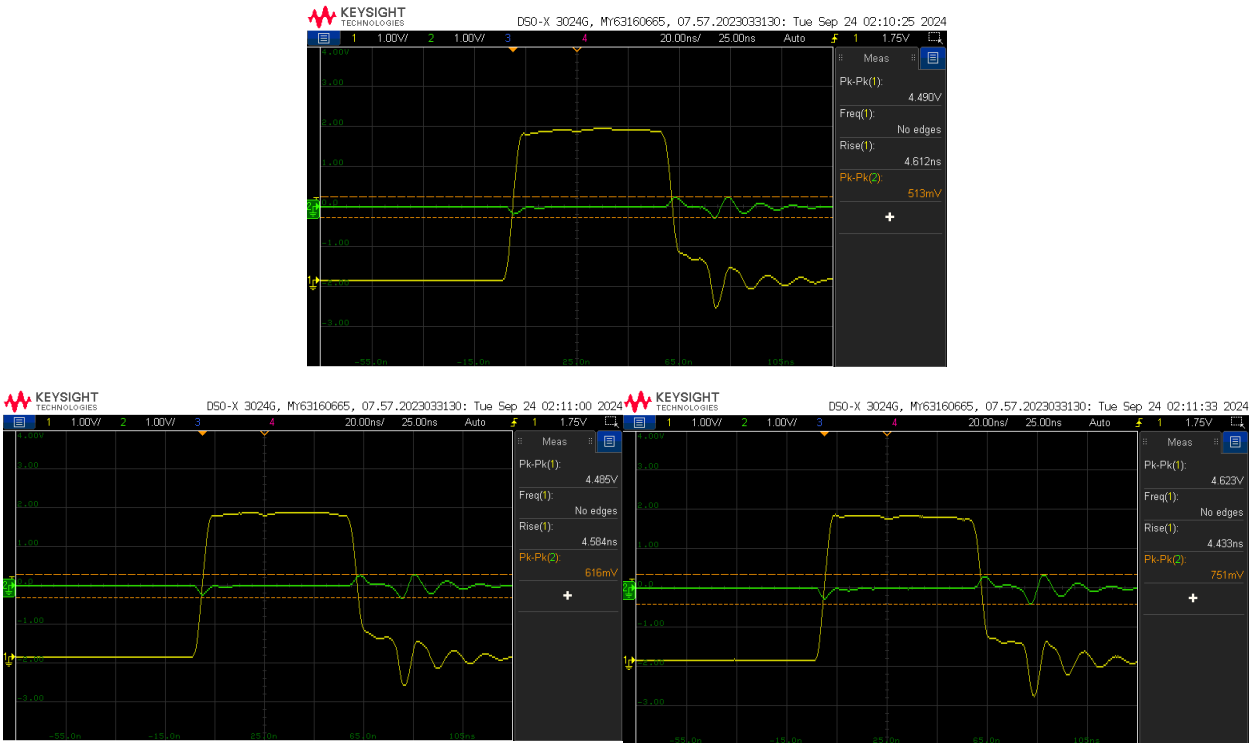


Fig 8: Switching noise when pins 8 and 9 (top) 8, 9 and 10 (bottom left) and 8, 9, 10 and 11 (bottom right) on a separate path

From figures 6, 7 and 8, we can deduce the behavior of the victim loop under different switching circumstances. From figure 7, we see that pin13 primarily dominates the noise on the victim loop since the addition of the other pins does not cause much noise. These findings imply that when there is a single return path, the further the trace is from a victim loop, the more cross-talk there is. Should there be a situation where using a separate return path is unavoidable, the traces should be kept closer together. This is likely due to the increased loop area between the aggressor and victim loops.

The noise on the victim loop is also affected by the amount of pins that are switching, as seen in figure 8. 8, 9, 10 and 11 caused the most noise on the victim loop.

One Pin Switching at a Time (Common Path):

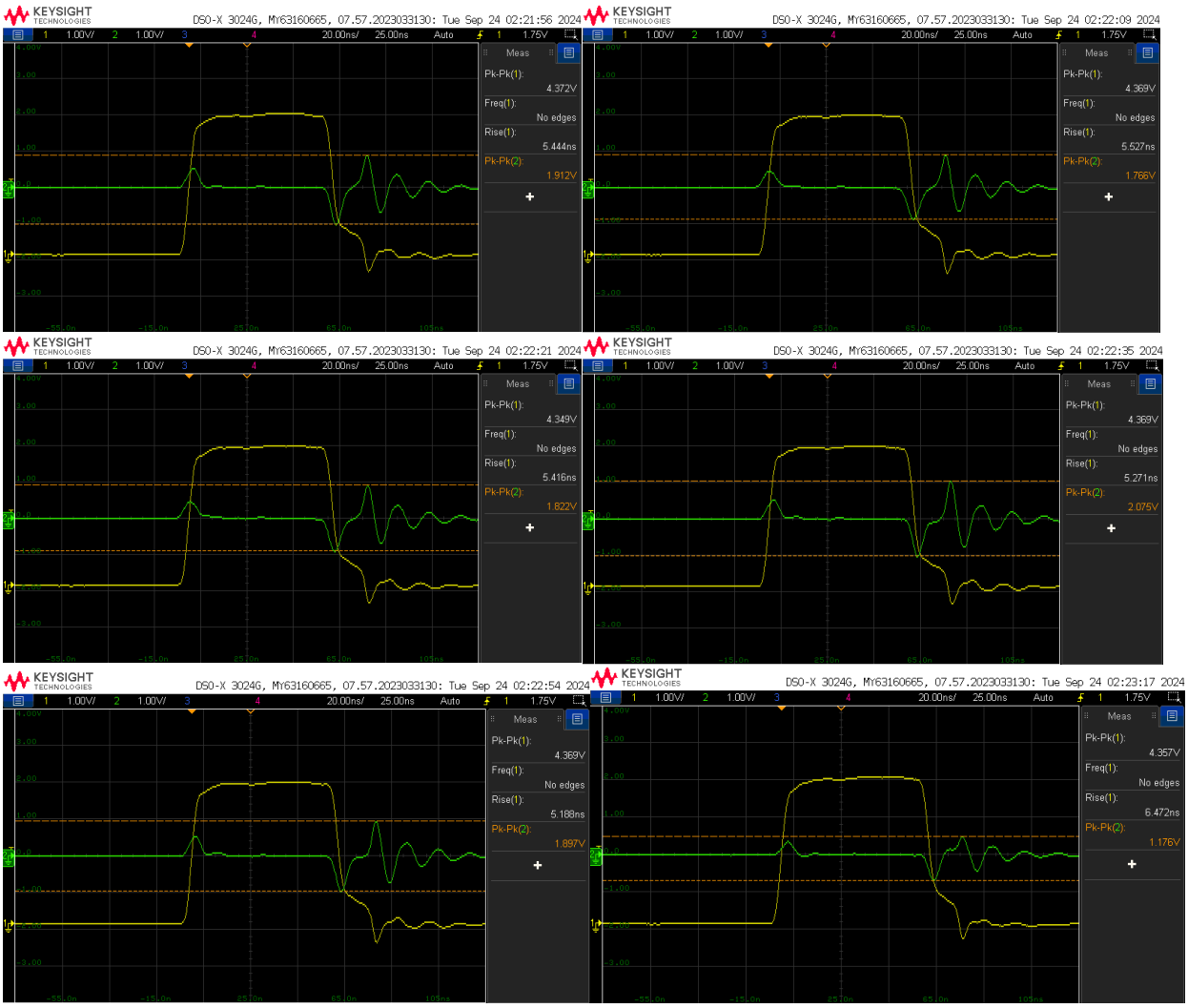
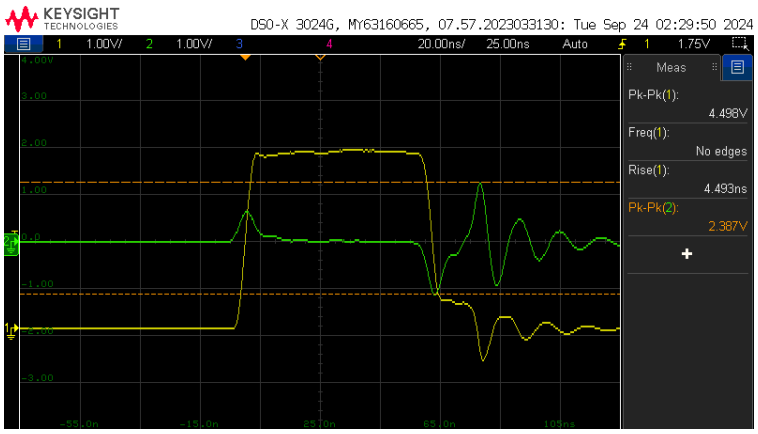
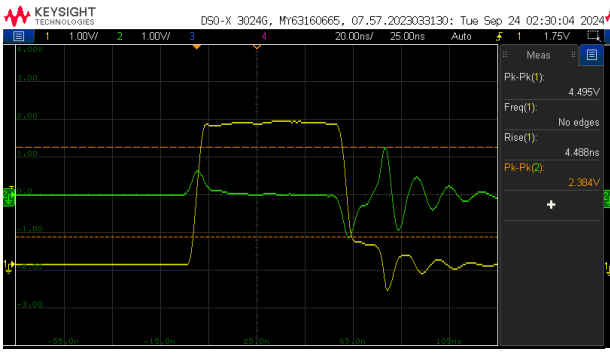


Figure 9: Switching noise on a common return path when pin 8 (top left), 9 (top right), 10 (middle left), 11 (middle right), 12 (bottom left) and 13 (bottom right)

Increasing Number of Pins Switching (Common Path):





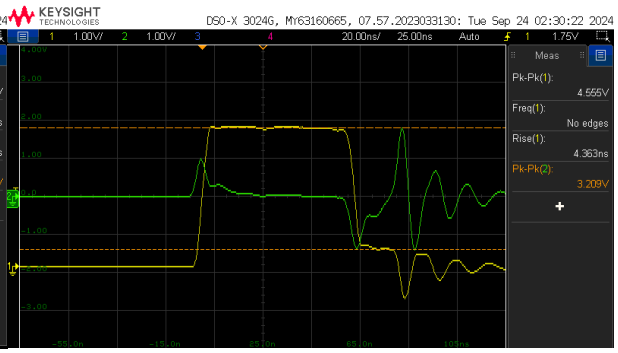


Fig 10: Switching noise when pins 8 and 9 (top) 8, 9 and 10 (bottom left) and 8, 9, 10 and 11 (bottom right) on a common path

From Figures 9 and 10, we see similar findings as in figure 6, that is, when the aggressor shares a return path with the victim, there is a voltage change on the same degree as the signal itself. Similar to figures 8 7 and 8, pin 13 contributes majorly to the noise seen on the victim loop, since there is a voltage change of 1.1v from just pin 13 itself.

A similar conclusion can be made from the common return path, rather than traces that are closer together, there is more noise induced by traces located further away from the victim loop due to the increase in area between the aggressor and victim. However, it should be noted that the inductance from the increase in area has a much greater effect if the loops share a return path rather than a separate path.

Conclusion and Key Takeaways:

The purpose of this lab was to solidify our understanding of return paths and why there is an emphasis on using a ground plane. Additionally, this lab which trace layout causes the most cross-talk between an aggress and victim loop. The following are key takeaways from this lab:

1. A common ground return plane should always be used where possible. Even in the worst case scenario, the cross-talk between the aggressor and victim is not as bad as a single return path.

2. Long, parallel signal traces in close proximity should be avoided where possible. Similar to the findings in lab 5, traces that are close will experience an induced inductance when there are fast switching signals.

3. Should there ever be a situation where a return path must be separate, using a shared return path must be avoided. The amount of voltage change experienced by a victim loop is on the same degree as the aggressor loop itself, which will change the way the board behaves.

4. In a situation where a separate return path must be used, it would be best to keep the traces closer together to mitigate the cross-talk as much as possible.

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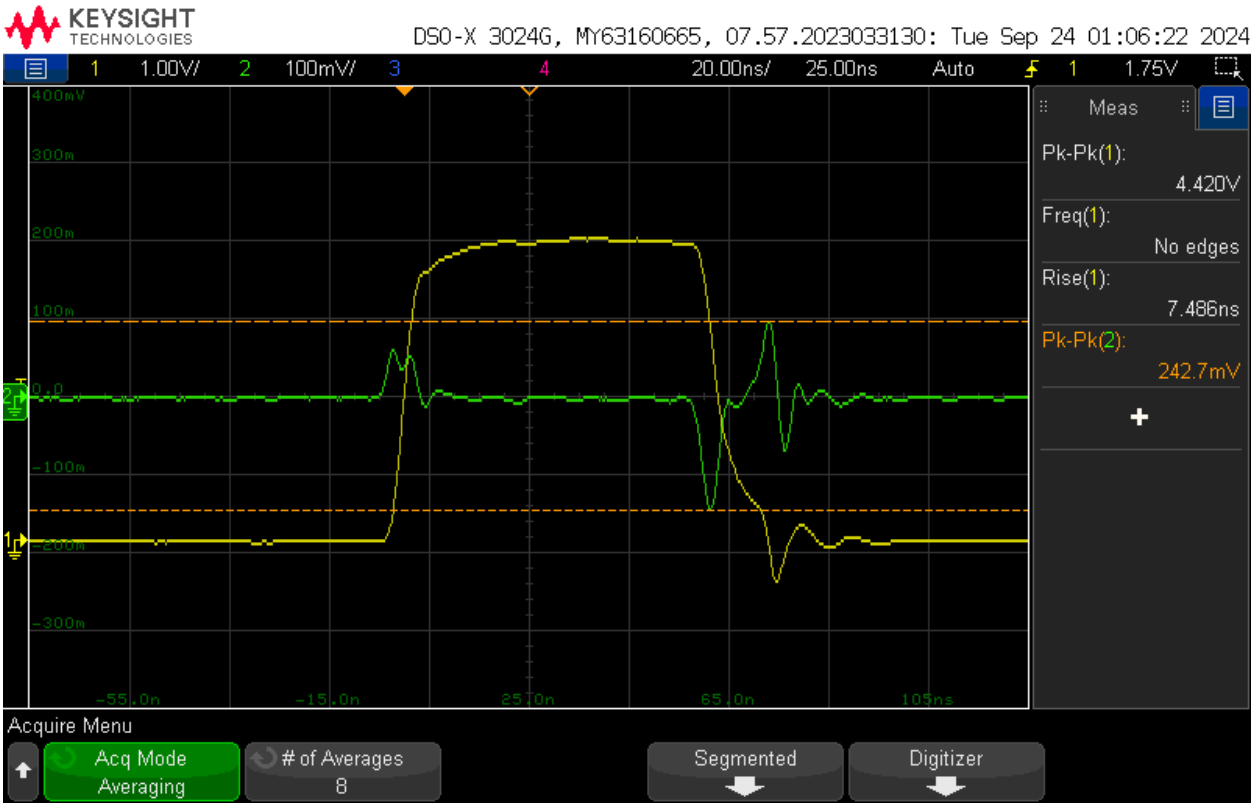
Part 1 Graphs:

One Pin Switching At a Time: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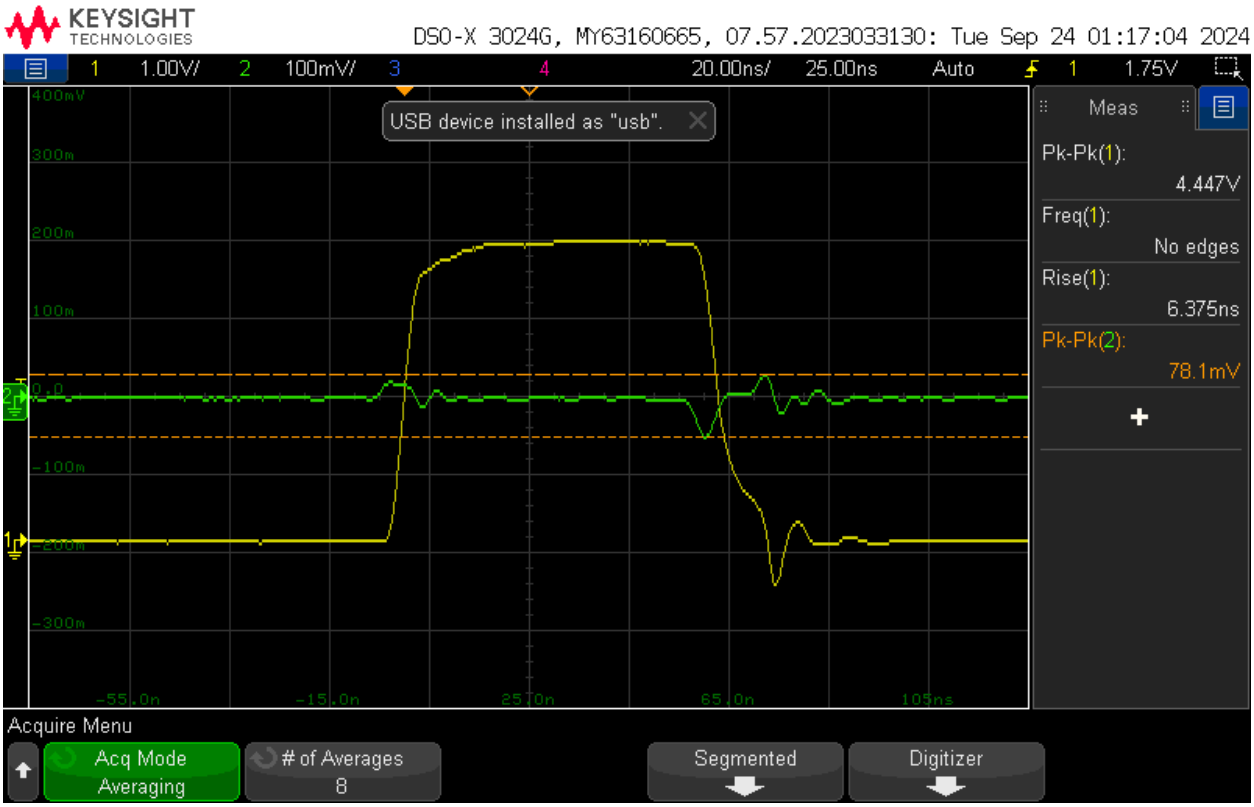
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 01:06:22 2024
1 1.00V/ 2 100mV/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1):
4.420V
Freq(1):
No edges
Rise(1):
7.486ns
Pk-Pk(2):
242.7mV
Acquire Menu
Acq Mode
Averaging
# of Averages
8
Segmented
Digitizer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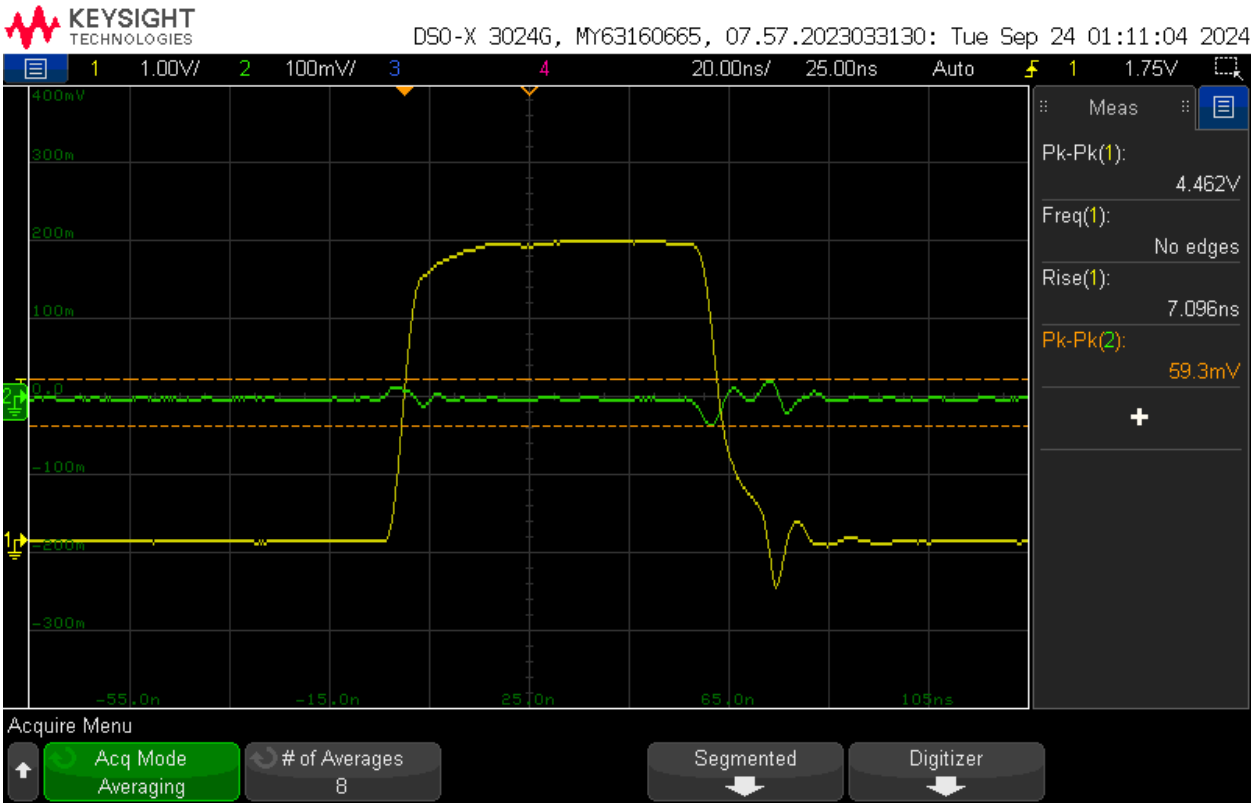
1 1.00V/ 2 100mV/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
USB device installed as "usb".
Meas
Pk-Pk(1):
4.447V
Freq(1):
No edges
Rise(1):
6.375ns
Pk-Pk(2):
78.1mV
Acquire Menu
Acq Mode
Averaging
# of Averages
8
Segmented
Digitizer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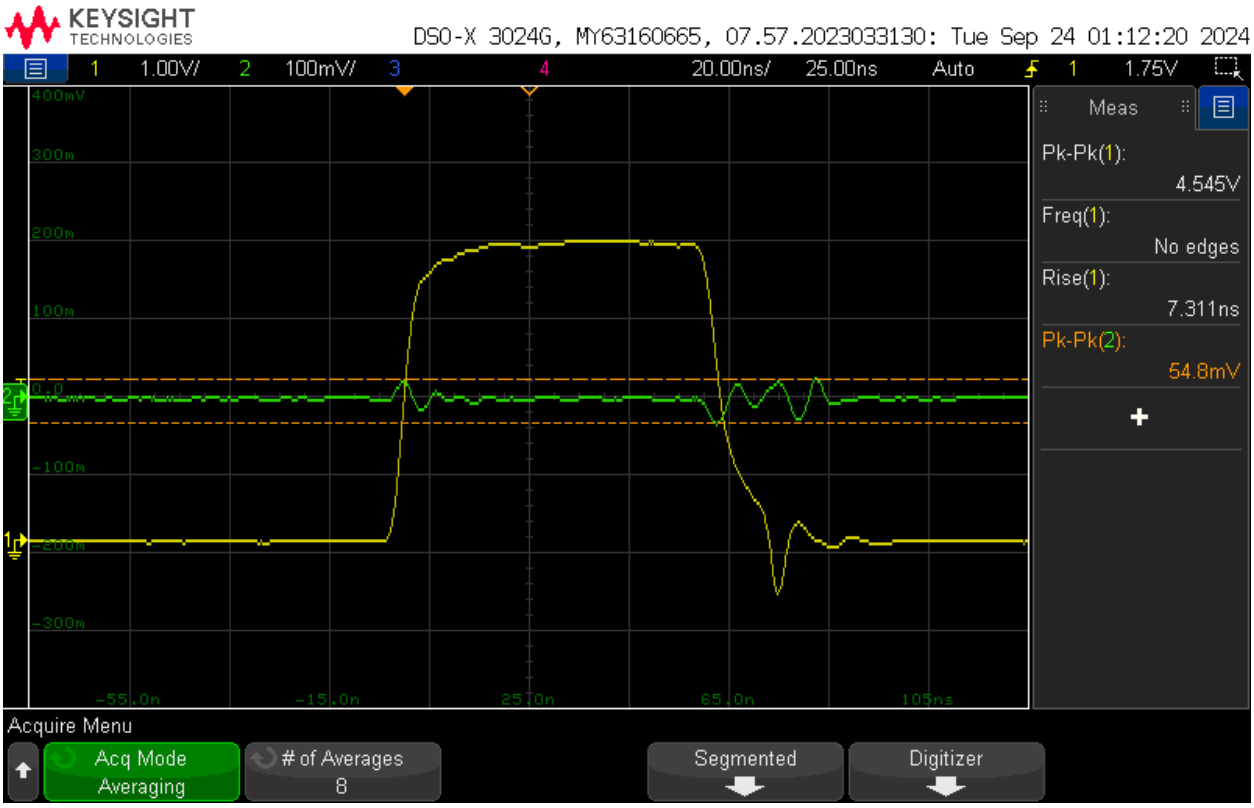
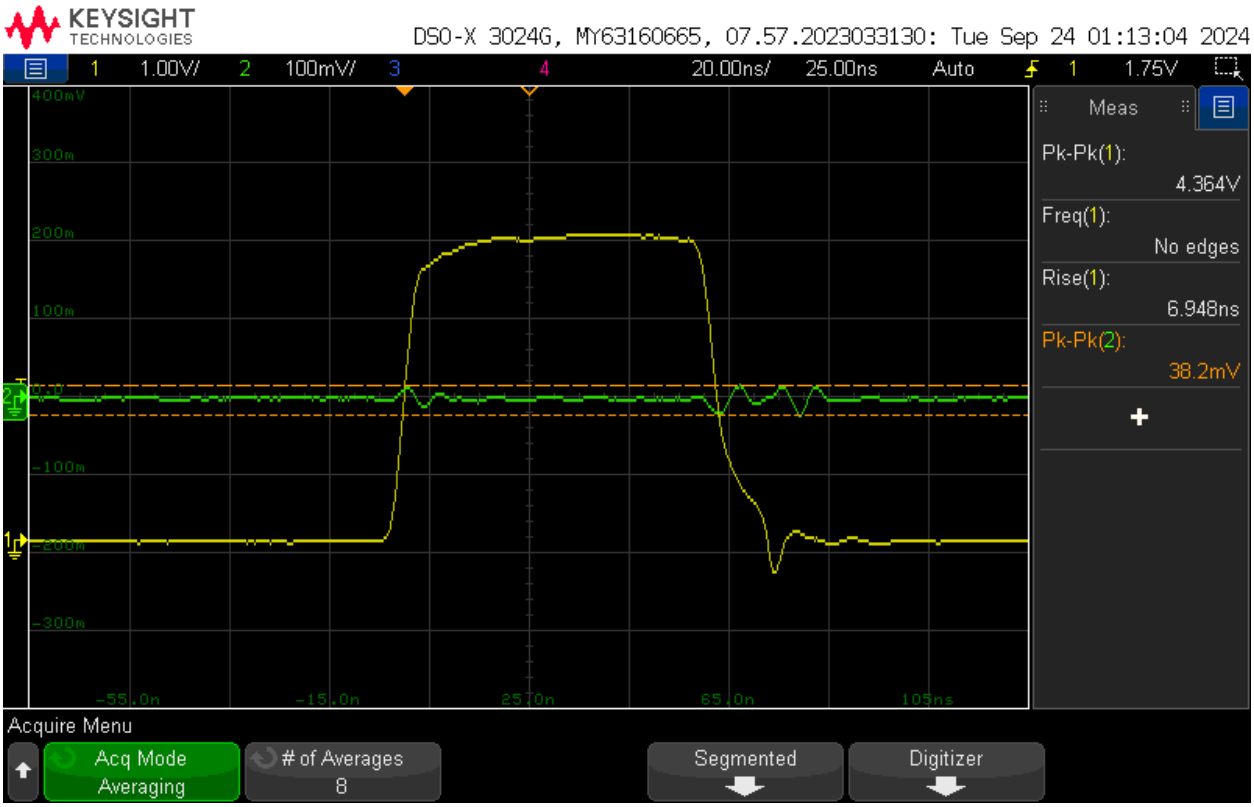
1 1.00V/ 2 100mV/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1):
4.462V
Freq(1):
No edges
Rise(1):
7.096ns
Pk-Pk(2):
59.3mV
Acquire Menu
Acq Mode
Averaging
# of Averages
8
Segmented
Digitizer

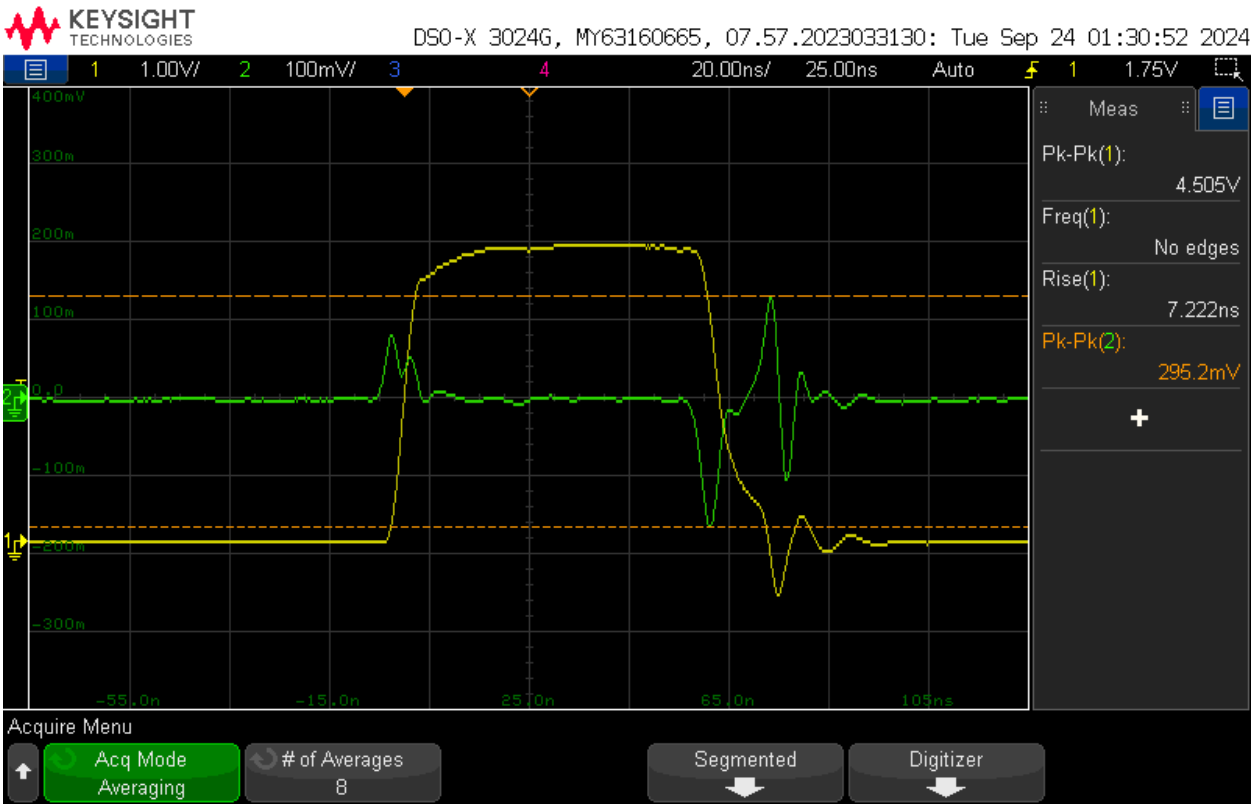
KEYSIGHT TECHNOLOGIES
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 01:11:45 2024
1 1.00V/ 2 100mV/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1):
4.447V
Freq(1):
No edges
Rise(1):
7.414ns
Pk-Pk(2):
65.8mV
Acquire Menu
Acq Mode
Averaging
# of Averages
8
Segmented
Digitizer

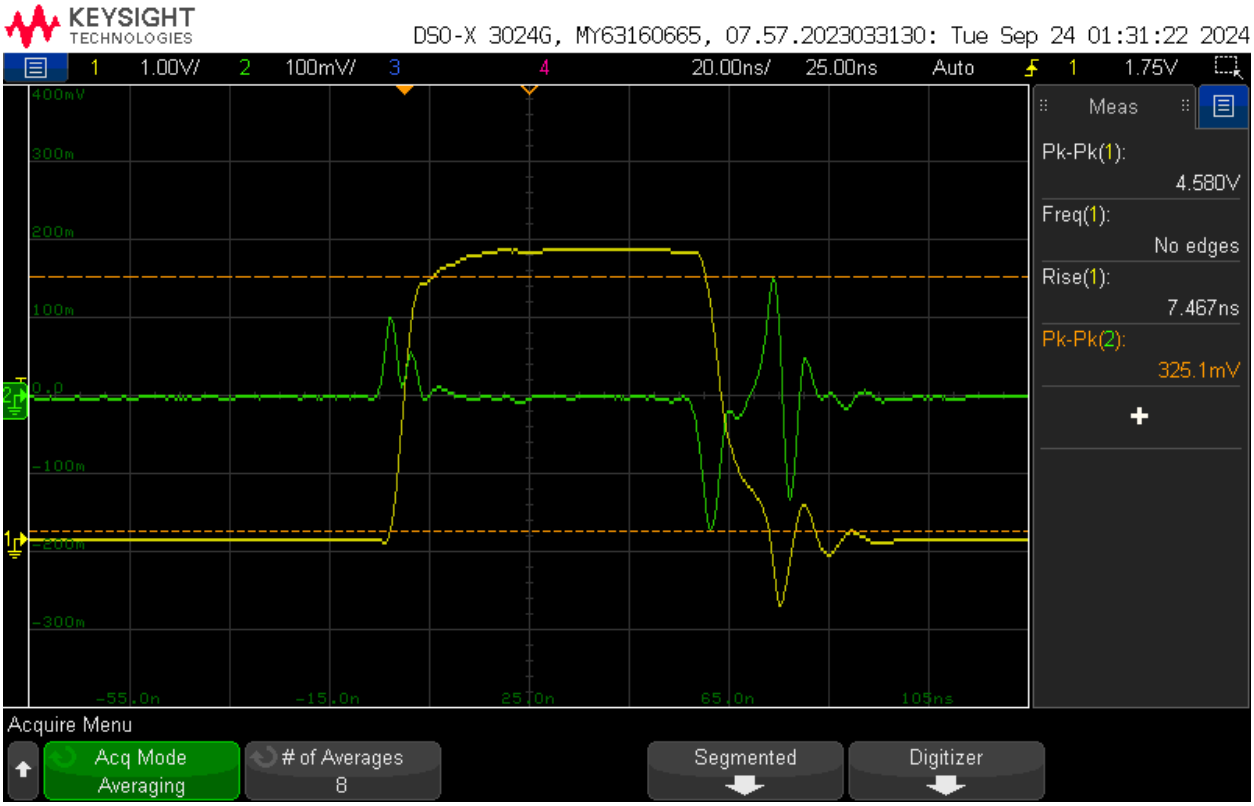


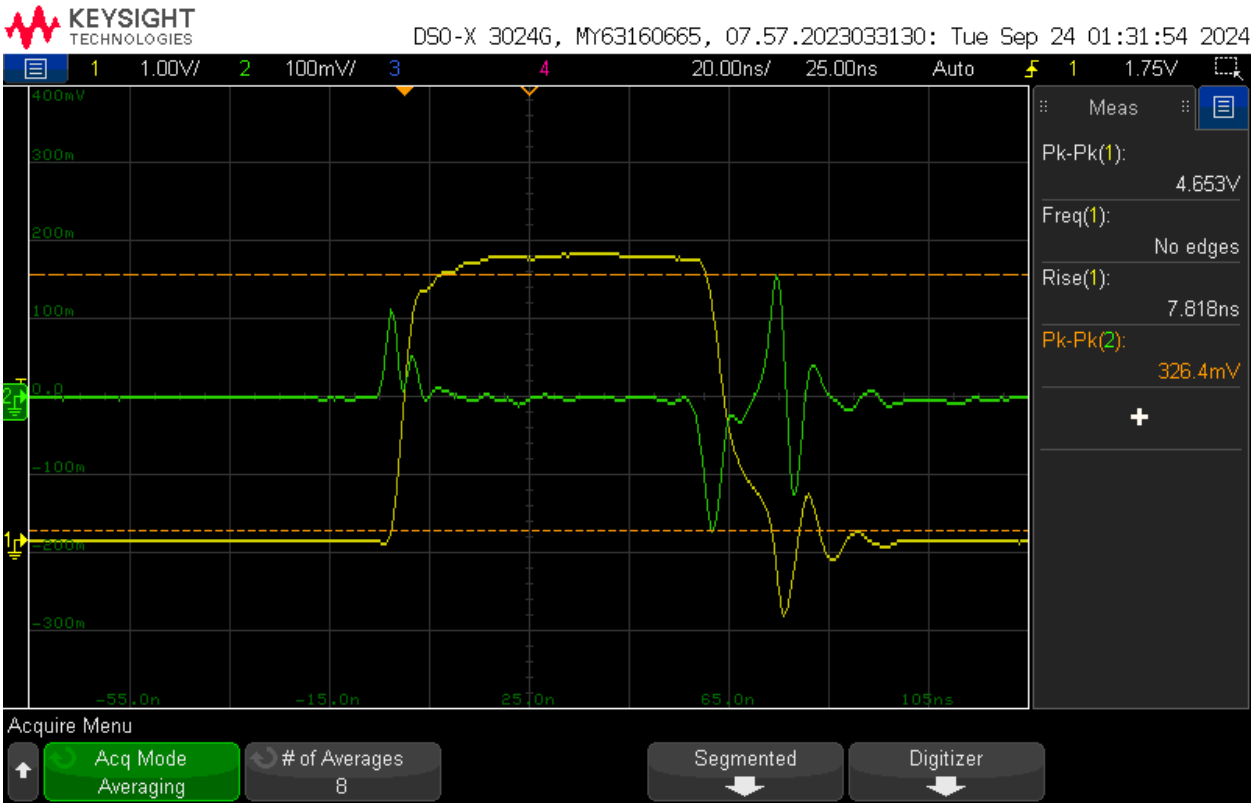


In descending order: Pin 8, 9, 10, 11 and 12 switching separately

Increasing Number of Pins Switching:



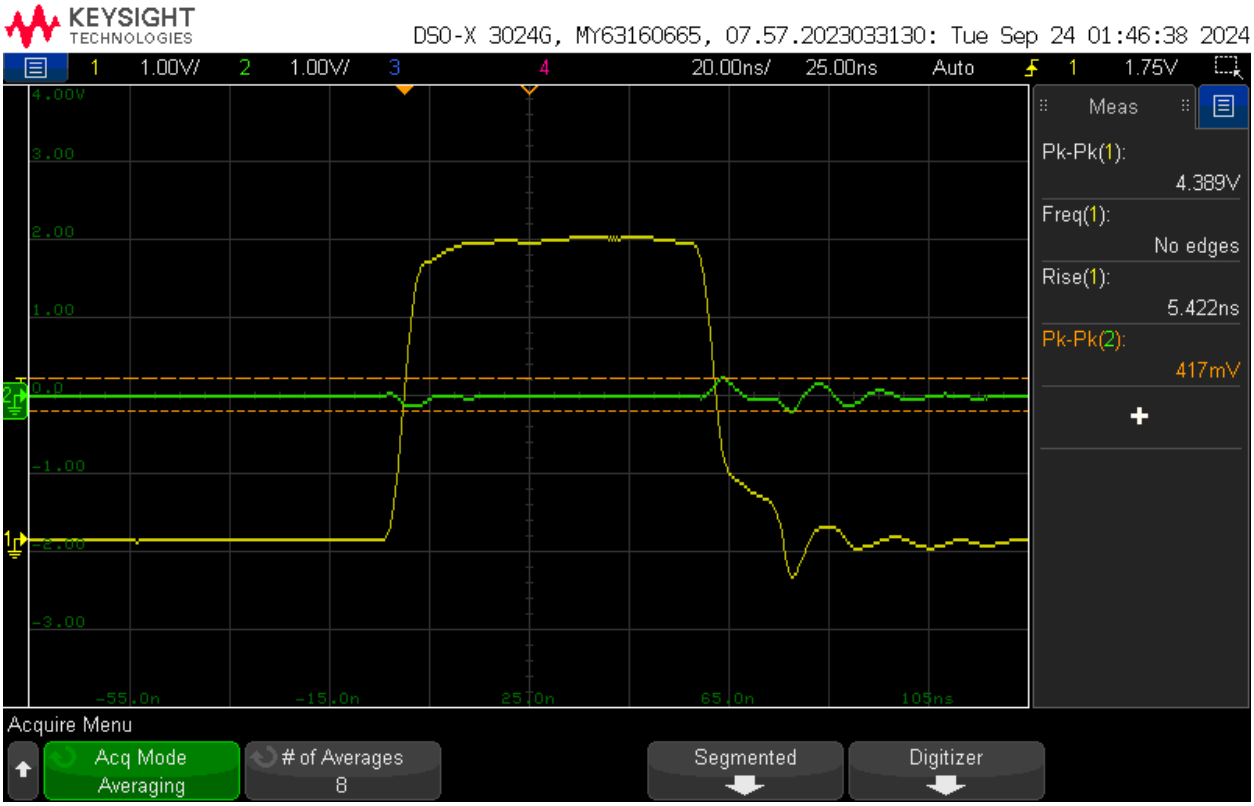


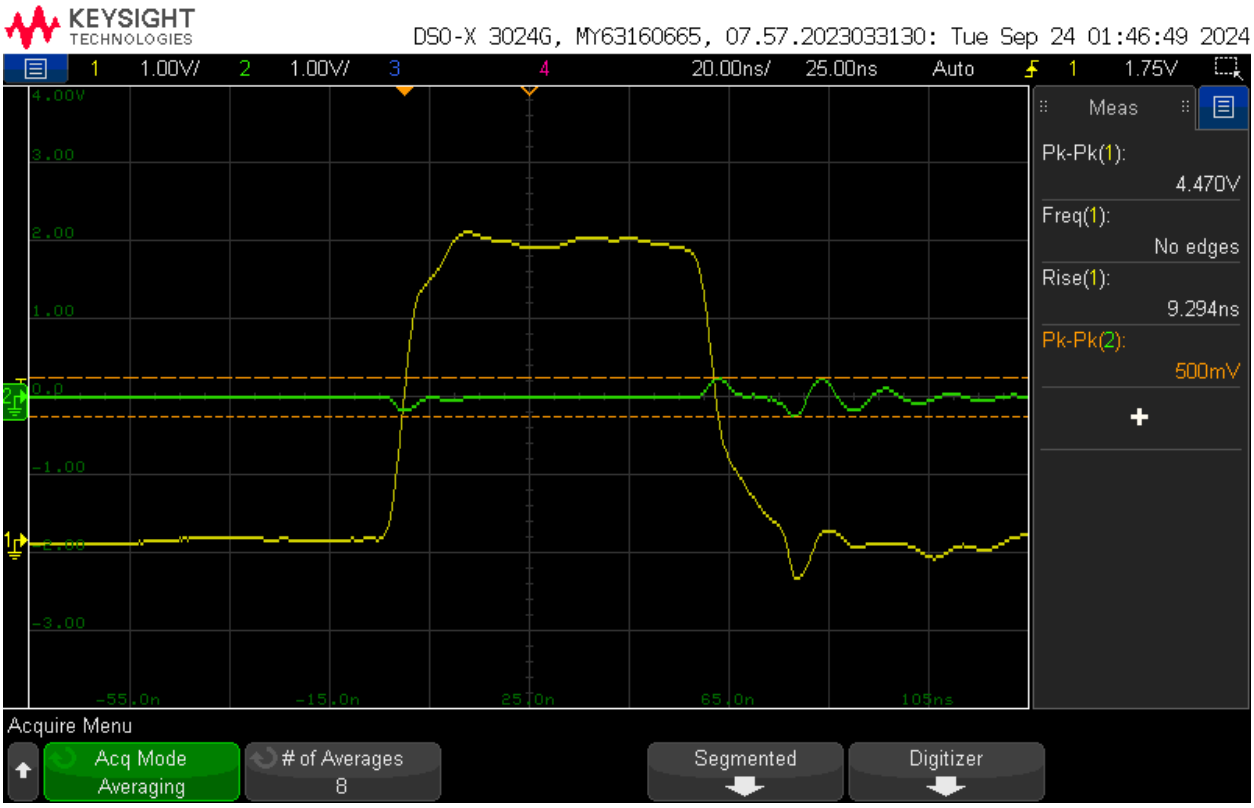


In descending order: Pins 8 and 9 | 8, 9 and 10 | 8, 9, 10, 11 | switching.

Part 2 Graphs:

One Pin Switching at a Time (Separate Paths)

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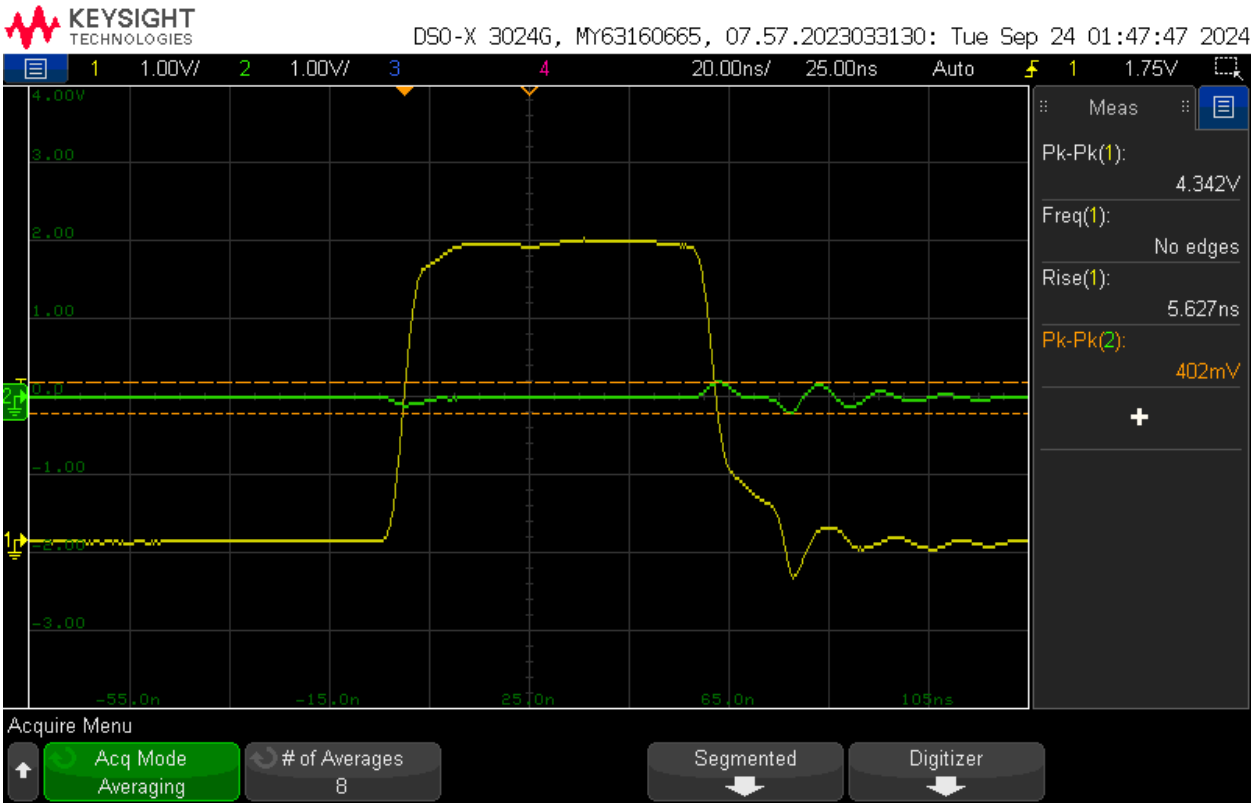
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 01:46:49 2024
1 1.00V/ 2 1.00V/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1): 4.470V
Freq(1): No edges
Rise(1): 9.294ns
Pk-Pk(2): 500mV
Acquire Menu
Acq Mode Averaging
# of Averages 8
Segmented
Digitizer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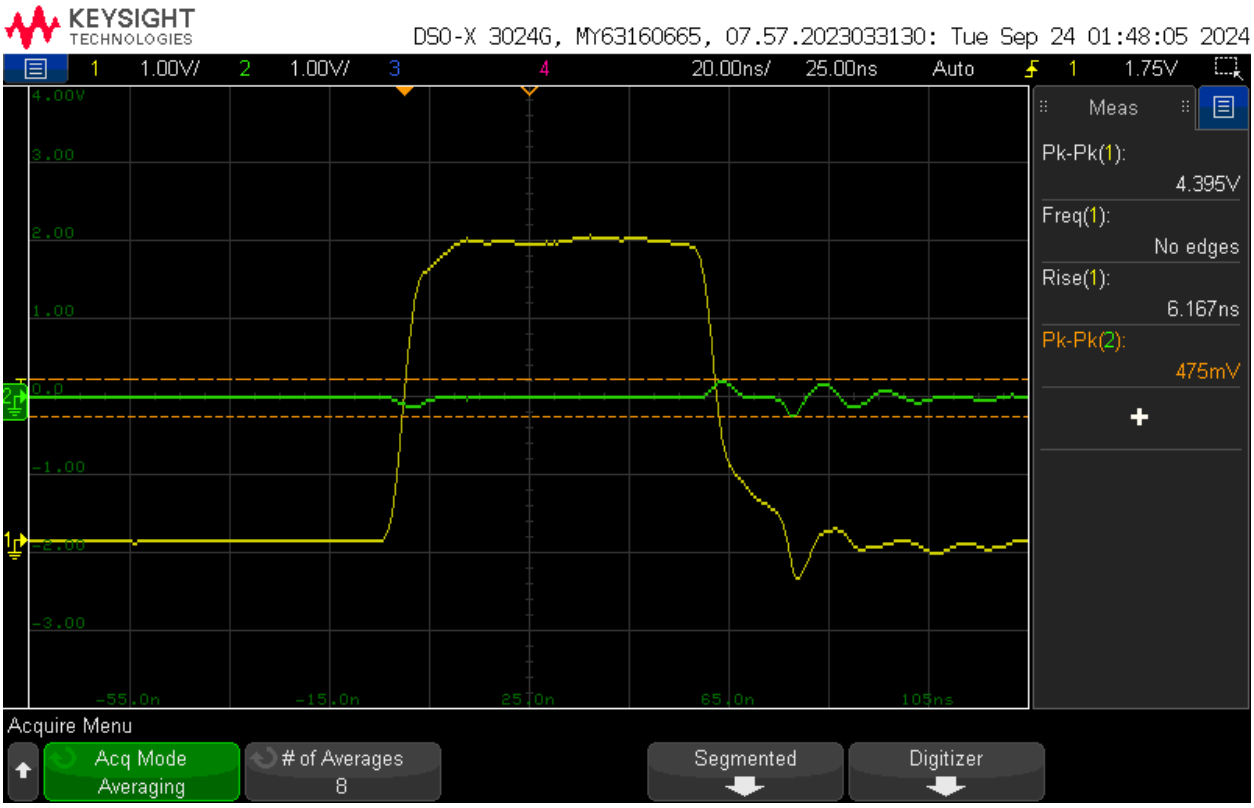
Meas
Pk-Pk(1): 4.342V
Freq(1): No edges
Rise(1): 5.627ns
Pk-Pk(2): 402mV
Acquire Menu
Acq Mode Averaging
# of Averages 8
Segmented
Digitizer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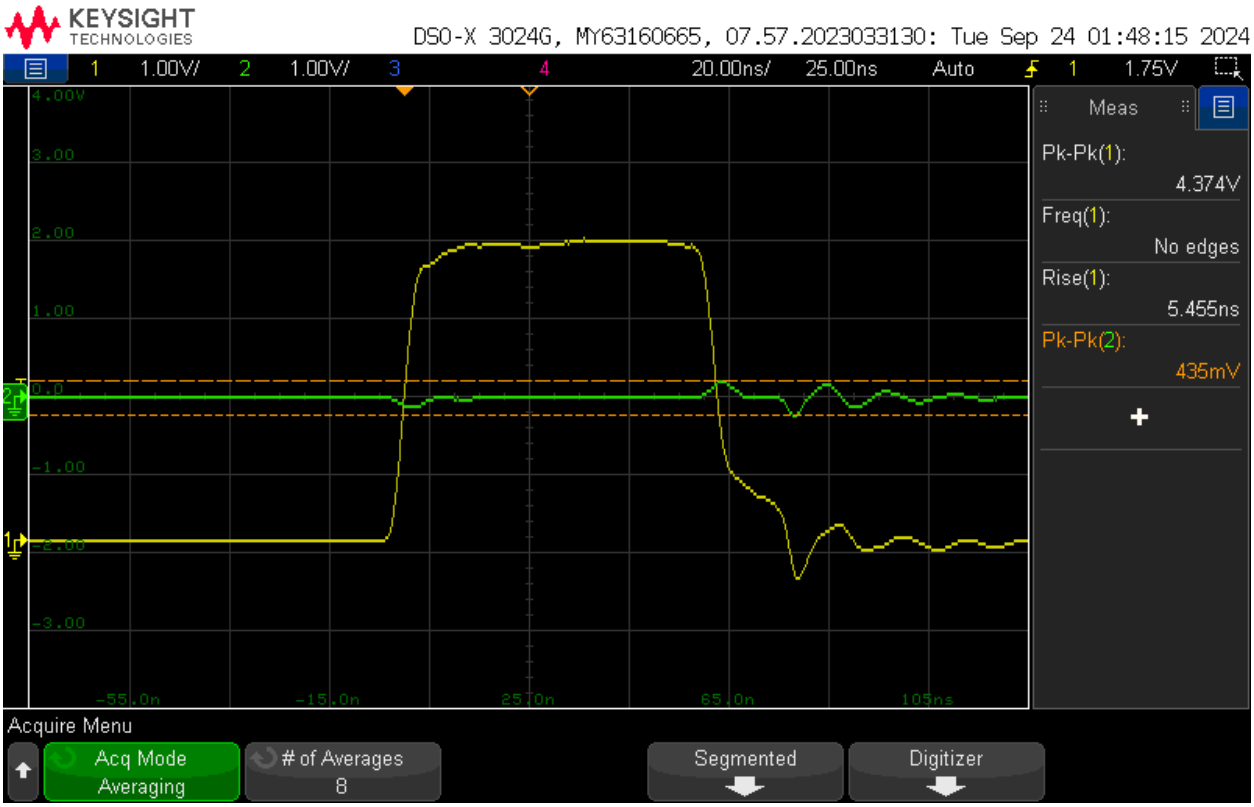
1 1.00V/ 2 1.00V/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1):
4.395V
Freq(1):
No edges
Rise(1):
6.167ns
Pk-Pk(2):
475mV
Acquire Menu
Acq Mode
Averaging
# of Averages
8
Segmented
Digitizer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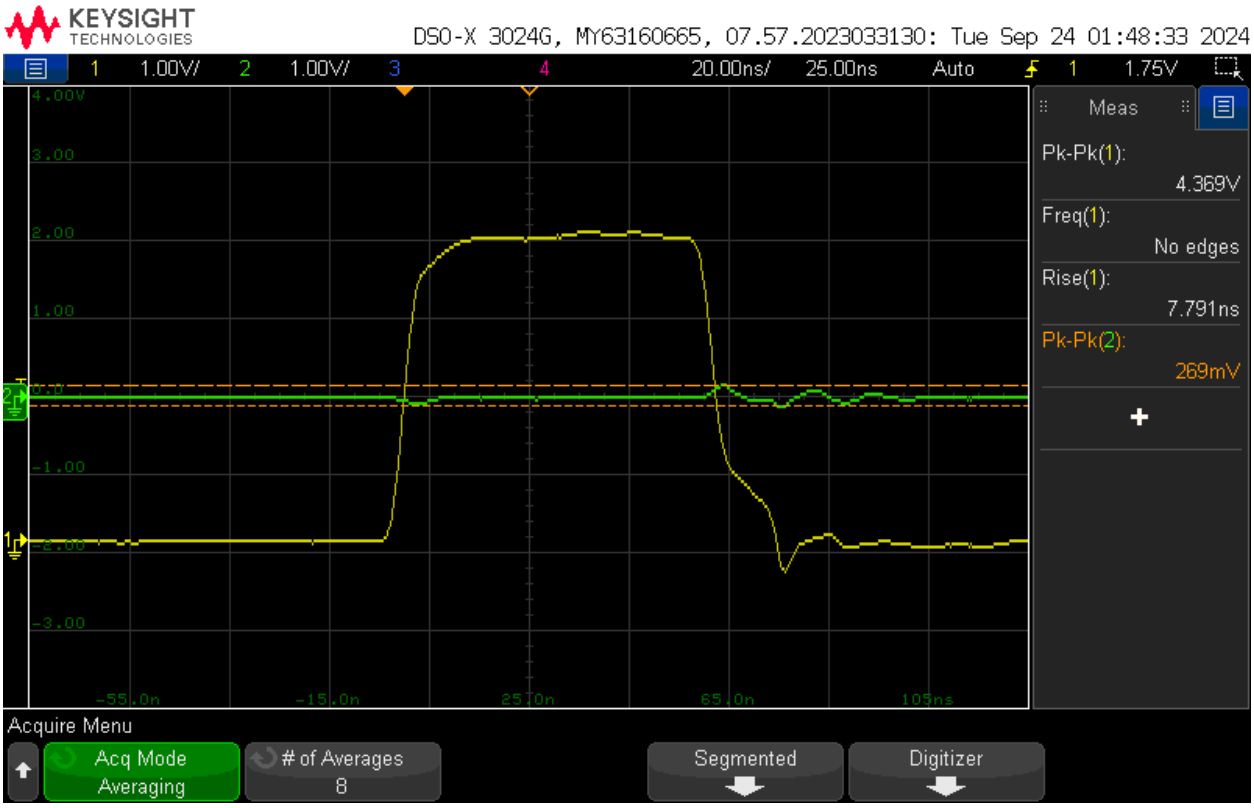
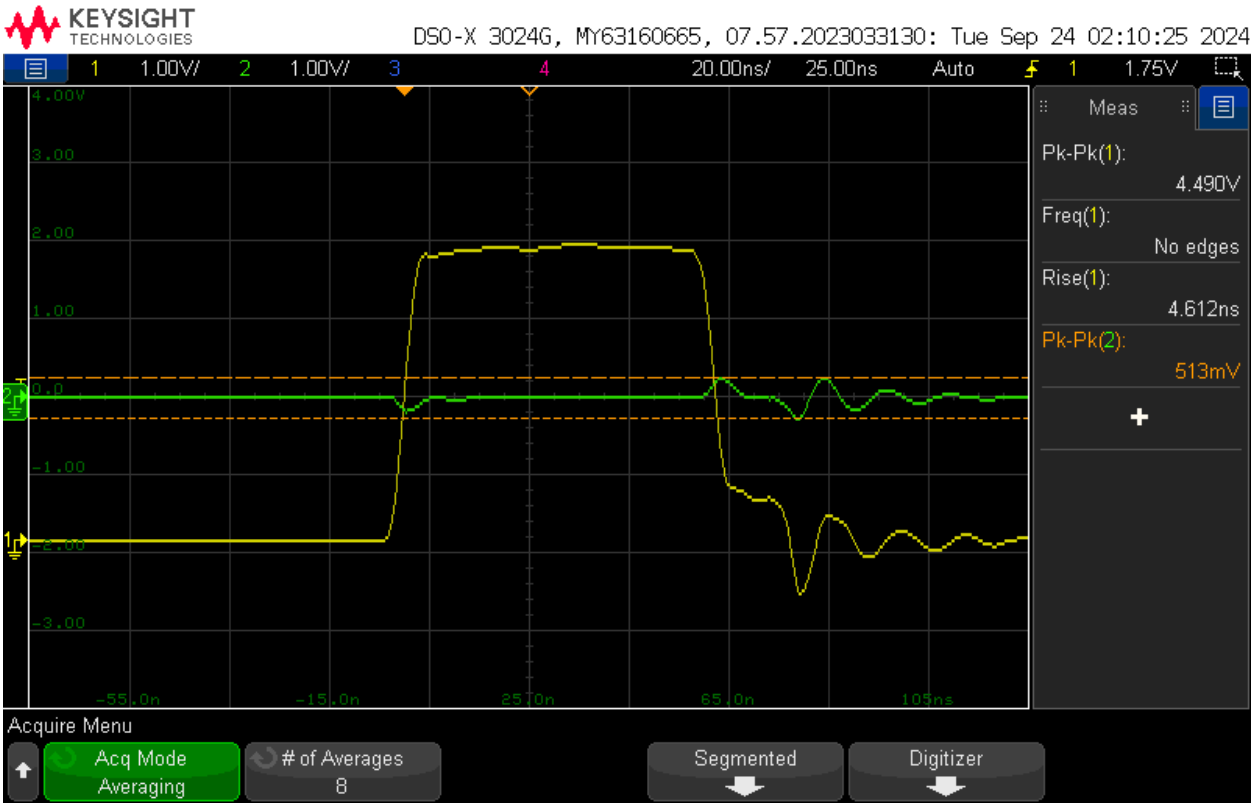
Meas
Pk-Pk(1):
4.374V
Freq(1):
No edges
Rise(1):
5.455ns
Pk-Pk(2):
435mV
Acquire Menu
Acq Mode
Averaging
# of Averages
8
Segmented
Digitizer



In descending order: Pin 8, 9, 10, 11 and 12 switching at a time

Increasing Number of Pins Switching (Separate Path)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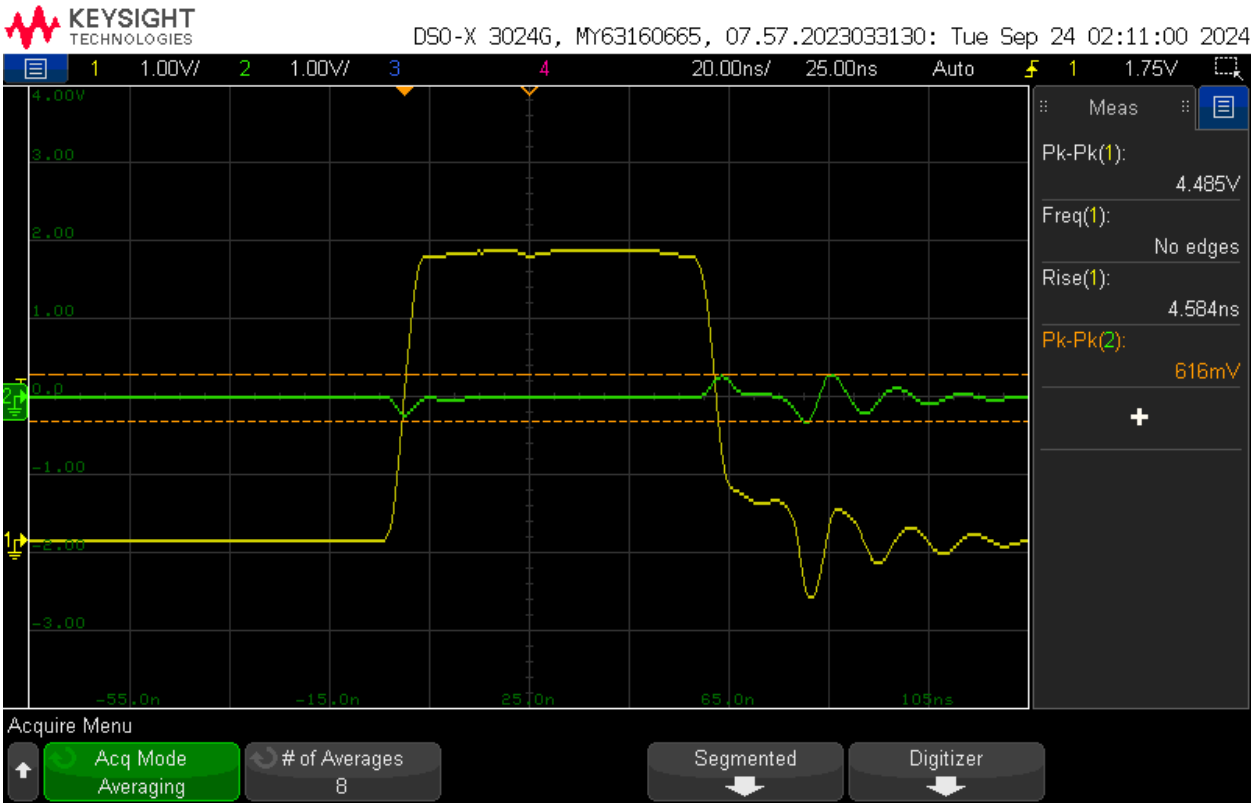
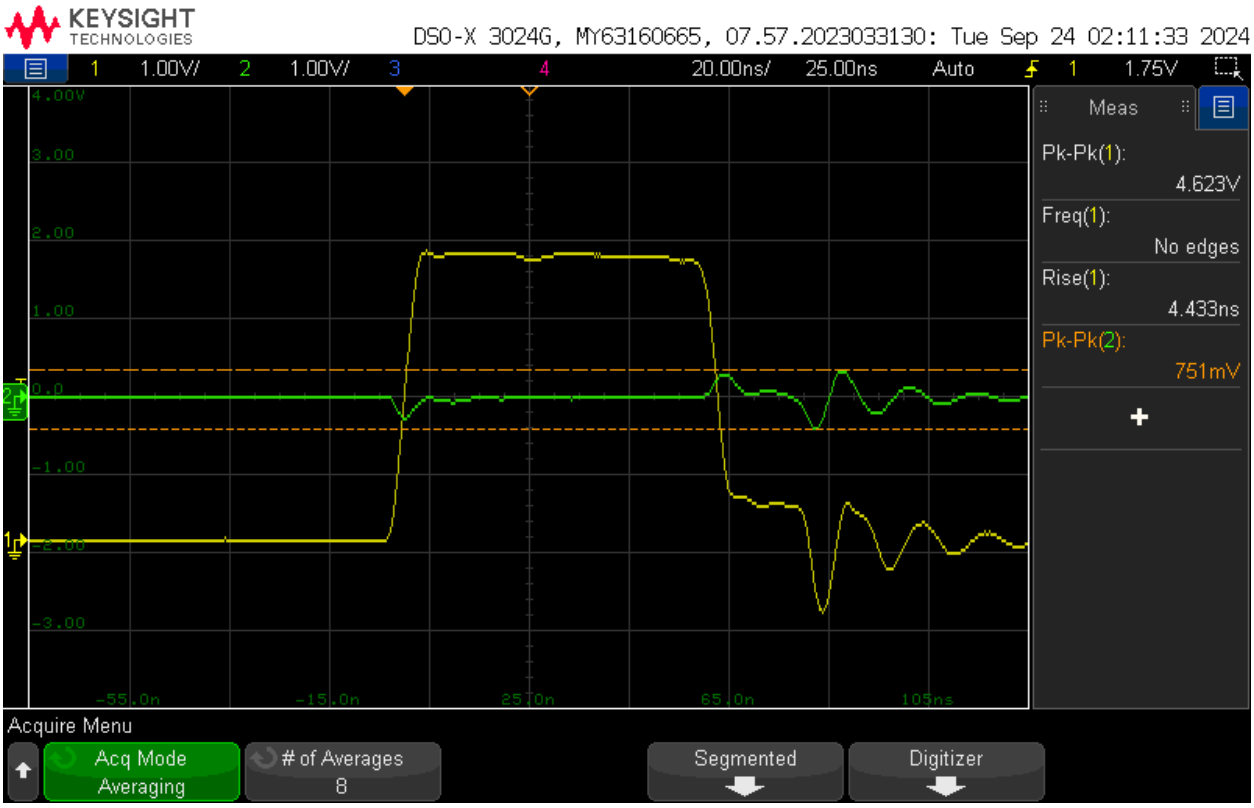
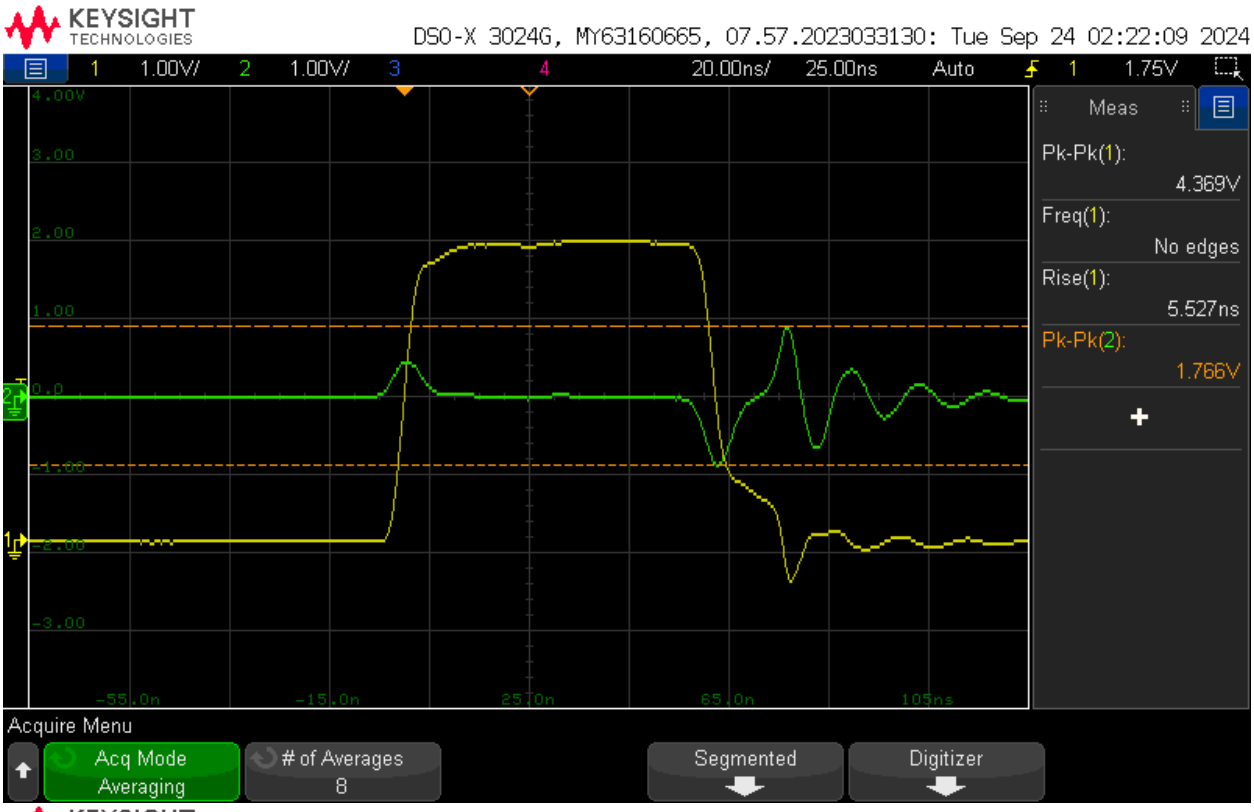
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 02:10:25 2024
1 1.00V/ 2 1.00V/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1):
4.490V
Freq(1):
No edges
Rise(1):
4.612ns
Pk-Pk(2):
513mV
Acquire Menu
Acq Mode
Averaging
# of Averages
8
Segmented
Digitizer

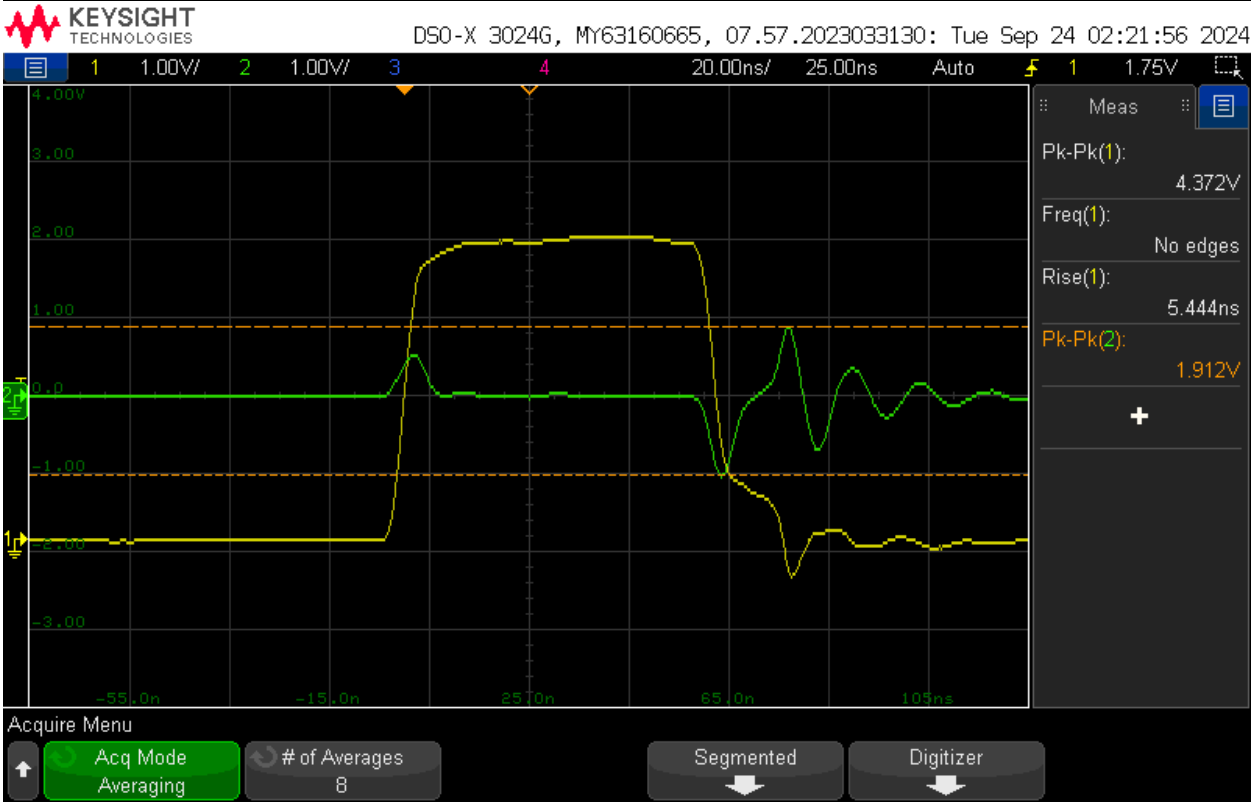

KEYSIGHT TECHNOLOGIES
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 02:11:00 2024
1 1.00V/ 2 1.00V/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1):
4.485V
Freq(1):
No edges
Rise(1):
4.584ns
Pk-Pk(2):
616mV
Acquire Menu
Acq Mode
Averaging
# of Averages
8
Segmented
Digitizer

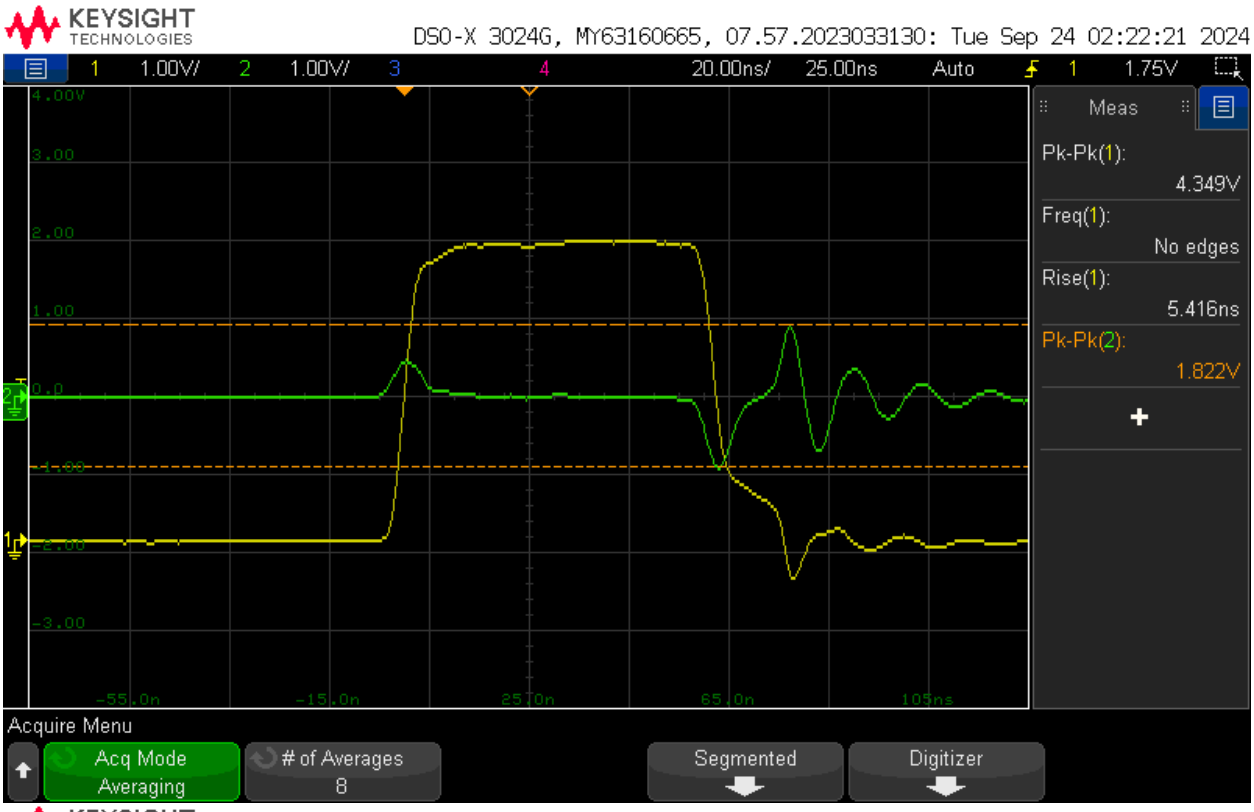


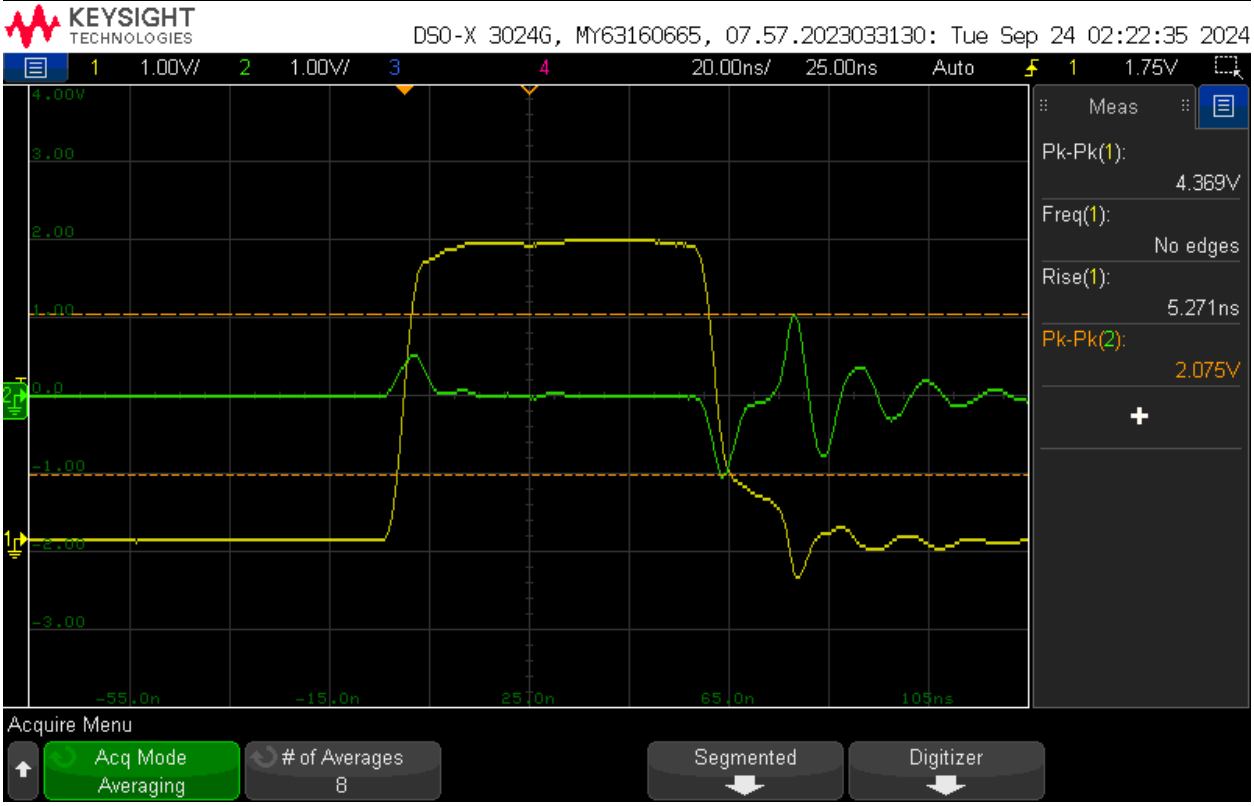
In descending order, Pins 8 and 9 | 8, 9 and 10 | 8, 9, 10 and 11 | switching

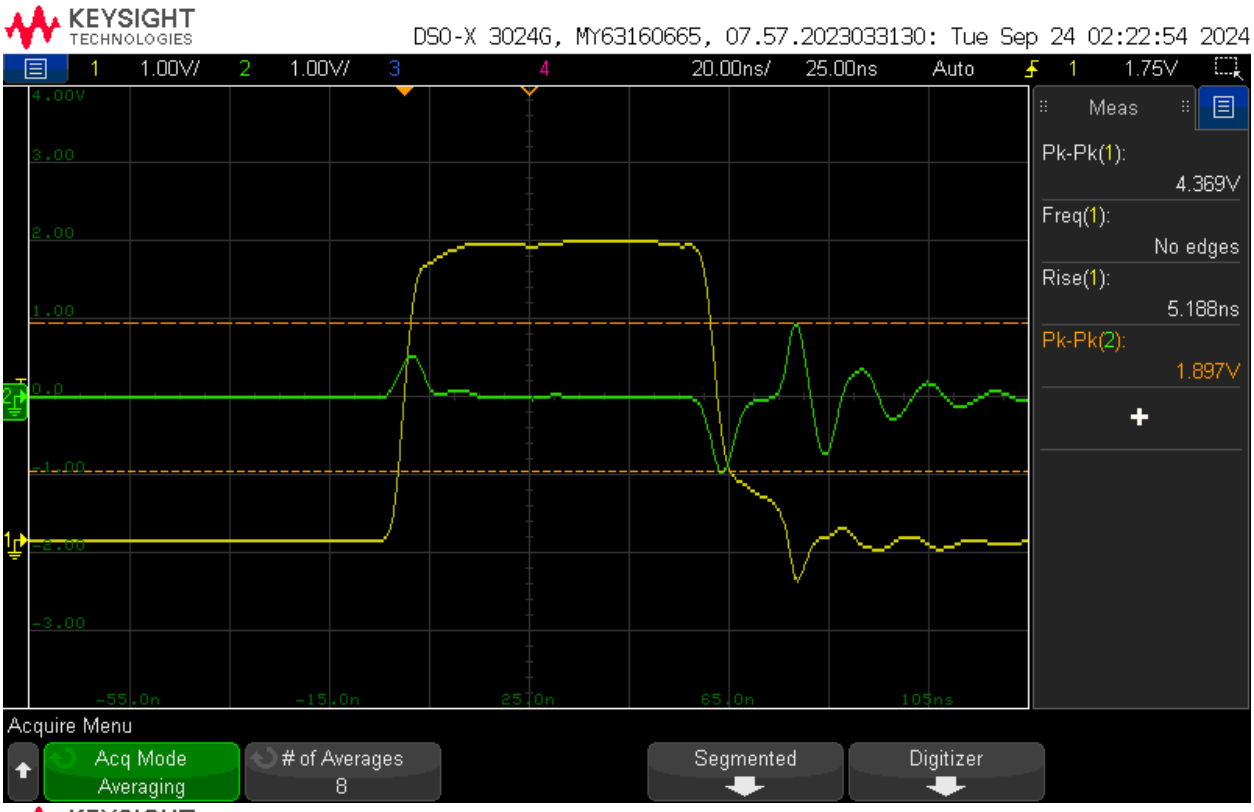
One Pin Switching at a Time (Common Paths)

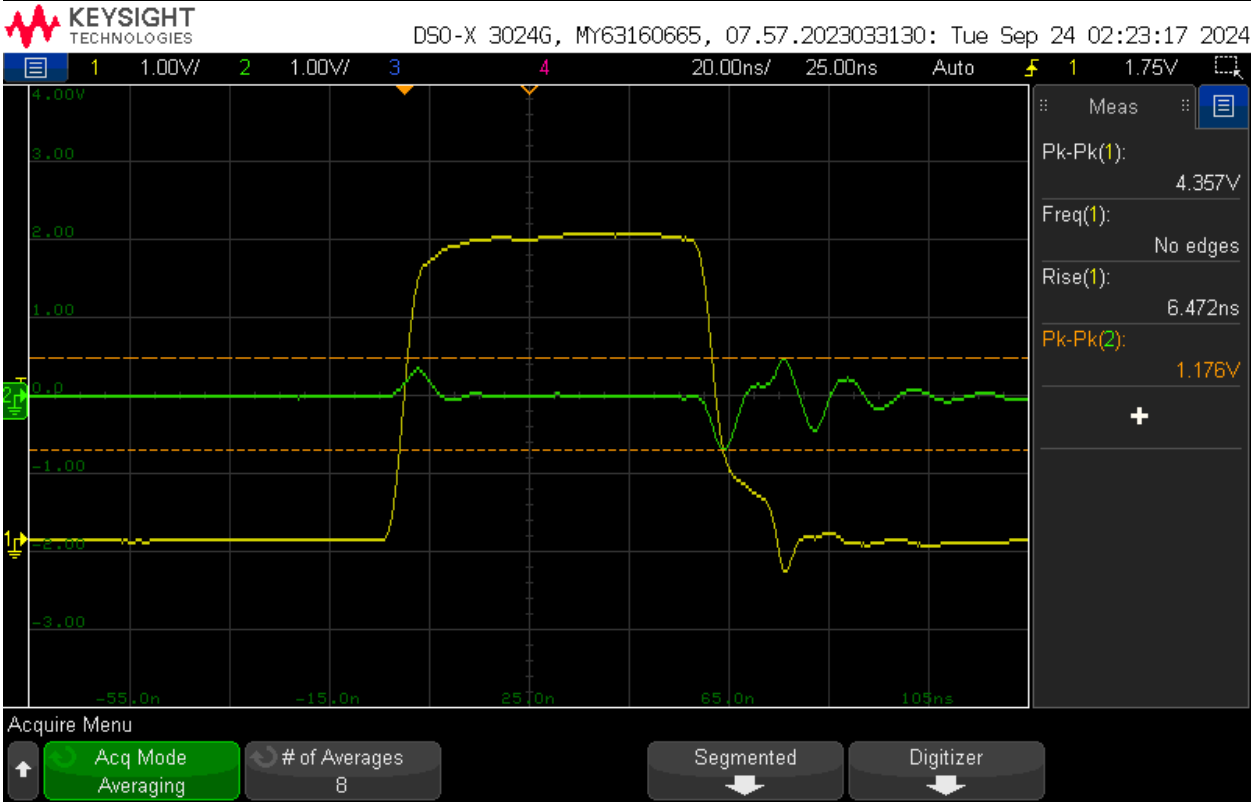

KEYSIGHT TECHNOLOGIES
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 02:22:09 2024
1 1.00V/ 2 1.00V/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1): 4.369V
Freq(1): No edges
Rise(1): 5.527ns
Pk-Pk(2): 1.766V
Acquire Menu
Acq Mode Averaging
# of Averages 8
Segmented
Digitizer


KEYSIGHT TECHNOLOGIES
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 02:21:56 2024
1 1.00V/ 2 1.00V/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1): 4.372V
Freq(1): No edges
Rise(1): 5.444ns
Pk-Pk(2): 1.912V
Acquire Menu
Acq Mode Averaging
# of Averages 8
Segmented
Digitizer


KEYSIGHT TECHNOLOGIES
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 02:22:21 2024
1 1.00V/ 2 1.00V/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1): 4.349V
Freq(1): No edges
Rise(1): 5.416ns
Pk-Pk(2): 1.822V
Acquire Menu
Acq Mode Averaging
# of Averages 8
Segmented
Digitizer

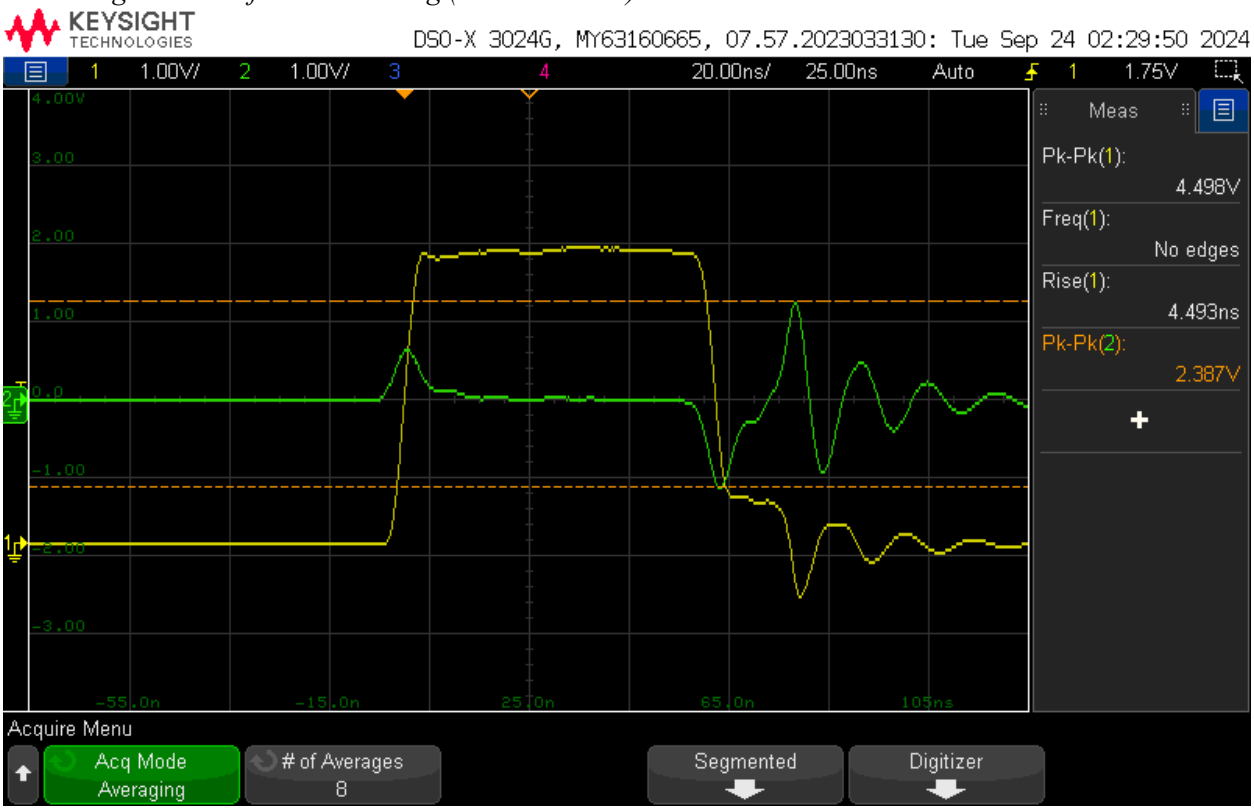

KEYSIGHT TECHNOLOGIES
DSO-X 3024G, MY63160665, 07.57.2023033130: Tue Sep 24 02:22:35 2024
1 1.00V/ 2 1.00V/ 3 4 20.00ns/ 25.00ns Auto 1 1.75V
Meas
Pk-Pk(1): 4.369V
Freq(1): No edges
Rise(1): 5.271ns
Pk-Pk(2): 2.075V
Acquire Menu
Acq Mode Averaging
# of Averages 8
Segmented
Digitizer

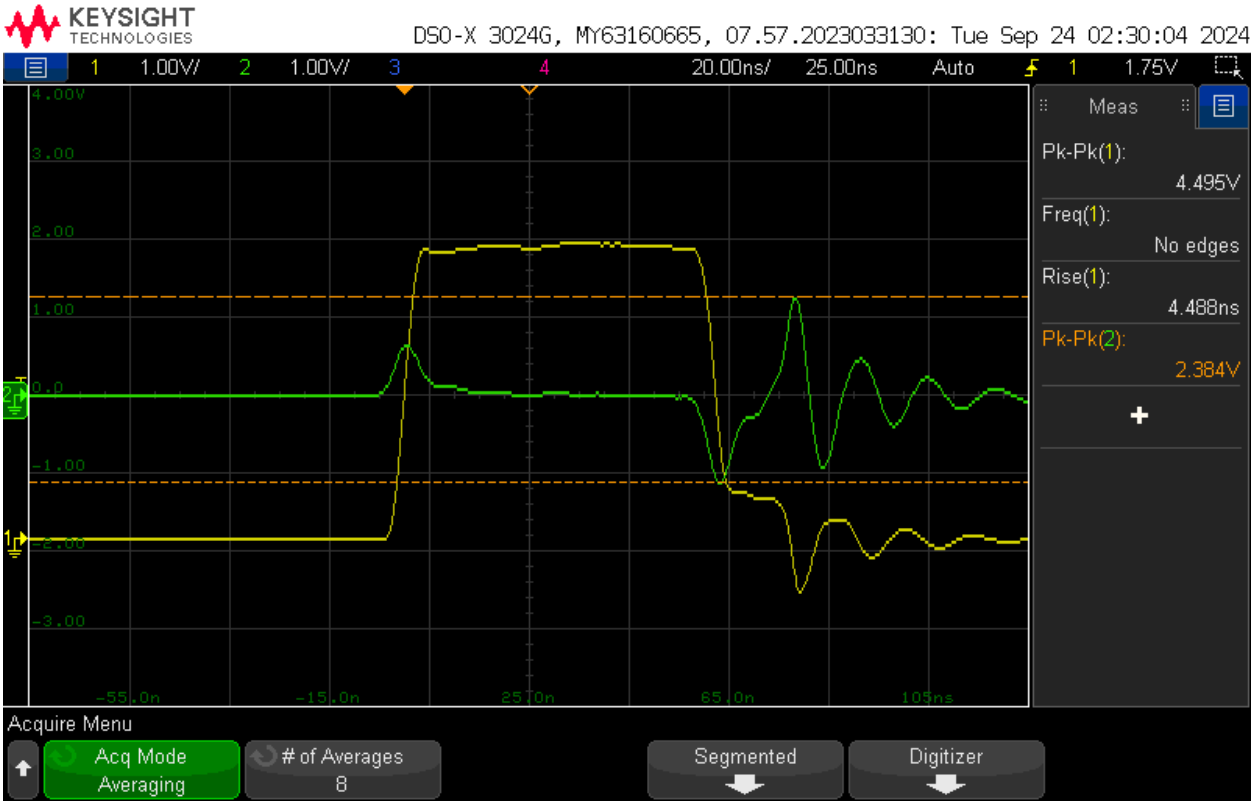


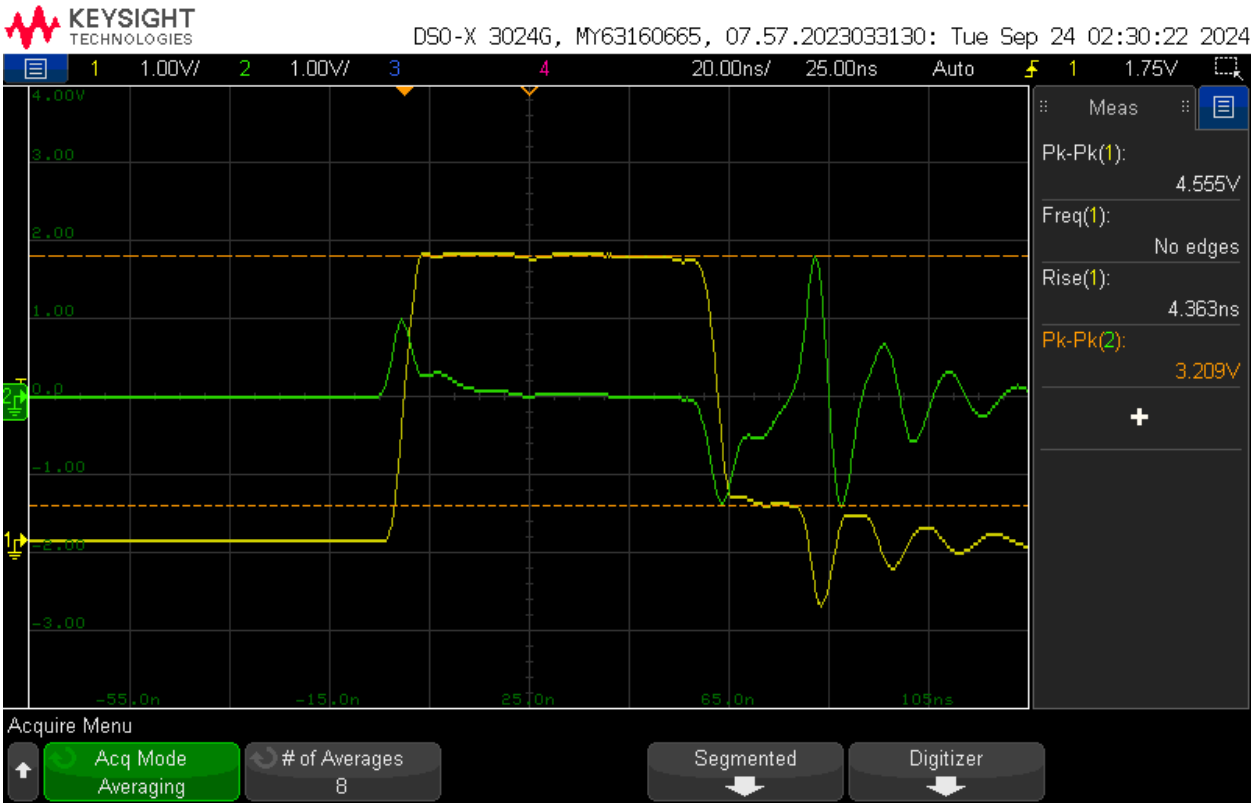


In descending order: Pin 8, 9, 10, 11 and 12

Increasing Number of Pins Switching (Common Path)







In descending order: Pins 8 and 9 | 8, 9 and 10 | 8, 9, 10 and 11 switching